PATHS

Grothendieck's path categories, simplicial localization,
and rectification of diagrams

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We work relative to Grothendieck universes $\mathbb{U} \in \mathbb{V}$. Unless explicitly enlarged, Cat denotes $\mathbb{U}$-small categories, $BX = |NX|$, and W_∞ denotes the functors whose nerves are weak homotopy equivalences. We write $\mathrm{Hot} = \mathrm{Cat}[W_\infty^{-1}]$. The category Cat hot has the same objects as Cat and functors modulo finite zigzags of natural transformations. An isomorphism in this category is stronger than a weak homotopy equivalence of nerves. No finiteness bound on the length of a zigzag may depend on an object of its source.

For a functor $u : A \to B$, the category A/b consists of pairs $(a, u(a) \to b)$ and $b\backslash A$ of pairs $(a, b \to u(a))$. We call u final if $b\backslash A$ is nonempty and connected for every b. We call it homotopy final if these categories are weakly contractible. A sieve is closed under sources; a cosieve is closed under targets. Coefficients in the manuscript are contravariant; when we write M^A we explicitly use covariant functors from A.

A fundamental localizer is a class W of functors satisfying weak saturation (identities, two-out-of-three and retracts), the terminal-object axiom $X \to e \in W$ when X has a terminal object, and the relative Theorem A axiom: for functors $A \xrightarrow{u} B \xrightarrow{v} C$, if $A/c \to B/c$ lies in W for every $c \in C$, then $u \in W$. A category is W-aspherical when its map to e belongs to W. This convention is used only where a localizer occurs; a relative category (M, Σ) initially requires just a wide subcategory $\Sigma \subset M$.

We distinguish equality, isomorphism in Cat, equivalence of categories, isomorphism in Cat hot, isomorphism in Hot, and a weak equivalence of a specified functor. A mapping space means a space in the ∞-categorical localization $L(M, \Sigma)$, with specified objects. The assertions below are labelled Proved, Refuted, Known, or Open; "Open" records what this investigation has not settled and makes no claim of research novelty. Prior AI-produced packages are evidence to check, not independent mathematical authorities. We claim no priority.

Manuscript page references use the supplied transcriptions of Grothendieck (1991). Abbreviations CATHOT, TOTASPH, POLES, GALOIS and STRATA identify the attached research drafts and reviews (Supplied AI research corpus, 2026); each invocation below states its independently checked scope. The research report records the passages read, computations, corrections and remaining questions.

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1 The path model and the total category

1.1 Endpoints, relative paths, and the component theorem

A word is read from its initial vertex to its terminal vertex, in the positional order of its vertices. A negative sign means that its generating arrow points against this positional order. It does not mean that the category M or its arrows have been replaced by an opposite category. We concatenate the word for a to b first and the word for b to c second, while writing the composite path as $d \circ c$.

The definition at VII ms. 65–67 calls the relative fixed-type category a "sous-catégorie pleine". Thus it restricts the backward arrows of objects, and does not restrict interior components of natural transformations. The text at ms. 66 saying that backward arrows "sont dans X" must read "sont dans Σ", as the supplied transcription already annotates. We use the precise two category constructions in B.1 throughout; the weak-vertical version is explicitly distinguished by its second superscript.

A.P1. The localization theorem on components PROVED. For a small category M, a wide subcategory Σ, and objects a, b, there is a natural bijection

$$\pi_0 \operatorname{Ch}_\Sigma(M; a, b) \cong \operatorname{Hom}_{M[\Sigma^{-1}]}(a, b).$$

It preserves concatenation and the empty-path identities. A backward edge is interpreted as the inverse of its arrow in the ordinary localization. No two-out-of-three, saturation, or model-category axiom is assumed.

Proof. Strict concatenation of the path categories defines a category $\Pi_\Sigma(M)$ with objects those of M and the displayed component sets as its Hom-sets. Sending an arrow to its one-edge positive path gives a functor $M \to \Pi_\Sigma(M)$. For composable arrows, the type map $(+) \to (+, +)$ gives a path morphism from their composite to their concatenation. For an identity arrow, collapse the single edge to the empty type. These observations prove the functor equations on components.

If $w : a \to b$ belongs to Σ, its reversed one-edge path is allowed. The two concatenations with that reversed path have types $(+, -)$ and $(-, +)$. In the first, the constant path maps to the backtrack by a transformation with middle component w; in the second, the backtrack maps to the constant path by such a transformation. Type collapse identifies each constant path with the empty path. Thus the image of w is invertible.

Let $F : M \to D$ be any functor sending Σ to isomorphisms. For a path c, define its effect in D by applying F, inverting the backward arrows, and composing. A type map does not change this effect. For a natural transformation of paths, its commuting squares give equal total effects because the endpoint components are identities. In checking a backward square one cancels the two invertible horizontal arrows; no vertical arrow is inverted. Hence the effect descends to a functor $\Pi_\Sigma(M) \to D$ extending F. It is unique because every path is the concatenation of its one-edge paths, and the backward ones have just been shown to be the prescribed inverses. This is the universal property of ordinary localization, with its structural functor from M retained. It proves the bijection and all its stated compatibilities. If a Hom-set is empty, both sides are empty by the same path description; disconnected categories present no additional condition. □

A.P2. The sequential models and the comparison roof PROVED. Let $t : \mathbb{N}^{\mathrm{op}} \to \mathrm{Ty}$ be a projective type sequence. With F_τ as in B.1, define

$$C_\infty(t; a, b) = \operatorname*{colim}_{n \in \mathbb{N}} F_{t_n}(a, b)$$

in Cat, with transition maps given by precomposition with $t_{n+1} \to t_n$. Let E_t be the fibred Grothendieck construction over $\mathbb{N}^{\mathrm{op}}$. There is a specified span

$$C_\infty(t; a, b) \longleftarrow E_t \longrightarrow \mathrm{Ch}_\Sigma(M; a, b).$$

The left leg sends a path to its colimit class. The right leg retains its type and path and applies t to a base arrow. The left leg is always in W_∞. For every sequence cofinal in the reachability preorder, both legs are in W_∞, by C.P3.

Proof. An arrow in E_t from (n, c) to (m, d), where $n \geq m$, is an arrow $c \to (t_n \to t_m)^* d$ in F_{t_n}. In the ordinary colimit this is an arrow $[c] \to [d]$, giving the left leg. Its type map and transformation give the right leg. Identities and compositions agree with the formula in B.1. The left weak-equivalence assertion is the filtered homotopy-colimit theorem in A.2; no full-faithfulness of transitions is required. The right assertion retains the exact additional hypothesis and refers to the independent enriched-type proof C.P1–C.P3.

For the block $D = (+, -)$, the left and right one-sided systems are represented by the two half-fences, and the symmetrically padded system by the integer fence J with edges $2k \to 2k-1$ and $2k \to 2k+1$. Paths are eventually constant on the indicated tails; fixed-endpoint transformations are eventually the specified endpoint identities. Constant-tail extension gives the actual q_L, q_R of CATHOT E.23–E.25. For a general block containing both signs, the corresponding models are the ordinary colimits of powers with left, right, or symmetric outer collapse. C.P4 proves the stated padding comparisons and their compatibility with composition for those precise transitions. No universal-arrow lifting property of the endpoint map is asserted by the term "path model"; Part E distinguishes it from smoothness. □

1.2 Total categories, ordinary colimits, and the missing cofinality hypothesis

A.1. The two Grothendieck constructions and their comparison maps PROVED. For a covariant diagram $D : A \to \mathrm{Cat}$, write $\int_A^{\mathrm{opf}} D$ for the category with objects (a, x), $x \in D(a)$, and arrows

$$(a, x) \longrightarrow (b, y), \qquad (u : a \to b,\ v : D(u)x \to y).$$

The superscript means the opfibrational construction. Its canonical functor

$$q_D : \int_A^{\mathrm{opf}} D \longrightarrow \mathrm{colim}_A^{\mathrm{Cat}} D$$

sends (a, x) to its colimit image and (u, v) to the image of v. For a strict contravariant diagram $F : T^{\mathrm{op}} \to \mathrm{Cat}$, write $\int_T^{\mathrm{fib}} F$ for the category whose arrows are $(u : t \to s, v : x \to F(u)y)$. These are the conventions at VII ms. 99–110. If $u : \mathbb{N}^{\mathrm{op}} \to T$ is a chosen sequence, $D = Fu^{\mathrm{op}} : \mathbb{N} \to \mathrm{Cat}$ and $C_u = \int_{\mathbb{N}^{\mathrm{op}}}^{\mathrm{fib}} D$, there are specified functors

$$C_u \xrightarrow{p} \int_T^{\mathrm{fib}} F = C, \qquad C_u \xrightarrow{q} \mathrm{colim}_n D(n) = C_\infty.$$

There need not be a functor $C \to C_\infty$.

Proof. The colimit identifies the two images of x and $D(u)x$, so the formula for q_D preserves identities and compositions. In the fibred case, an arrow $(n, x) \to (m, y)$ has $n \geq m$ and an arrow $x \to D(m \leq n)y$ in $D(n)$; its image is the corresponding arrow $[x] \to [y]$ in the colimit. The functor p is identity on fibre data and applies u to the base arrow. The formulas type every arrow in the display. The manuscript correctly uses precisely this span at ms. 108; inverting q will give $C_\infty \to C$ in Hot, rather than an otherwise unspecified map in the reverse direction. □

A.2. Filtered category colimits PROVED. If A is a small filtered category, then q_D of A.1 belongs to W_∞ for every $D : A \to \mathrm{Cat}$, with no injectivity, full-faithfulness, or fibrancy assumptions on its transition functors. In particular $q : C_u \to C_\infty$ is in W_∞ for every sequence. The same universal assertion holds if each connected component of A is filtered, and also for the empty category.

Proof. Fix an object z of the category colimit. An object of q_D/z is an object $x \in D(a)$ together with an arrow $[x] \to z$ in the colimit. Every finite list of objects, arrows, and equalities in a filtered category colimit is represented at a common stage. For completeness, construct a filtered category colimit by taking filtered colimits of the object and arrow sets. Source, target, identities, and composition descend because filtered colimits of sets commute with finite limits; an equality of two represented arrows holds at a further stage. The resulting category has the requisite universal property.

The category q_D/z is nonempty by representing z at some stage. Given a finite diagram in q_D/z, choose a common stage b representing its objects, all its arrows to z, and all its commuting triangles. Choose $z_b \in D(b)$ representing z. After passing to a further stage if necessary, the arrows to z_b give a cocone from that finite diagram to (b, z_b, id_z) in q_D/z. Thus q_D/z is filtered. Its nerve is weakly contractible: the image of any finite simplicial set in its nerve admits a cone after collecting its finite data, so each sphere map becomes nullhomotopic. Quillen's Theorem A (Lurie, 2026, Example 7.2.3.3, applied also to opposites) makes the specified q_D a nerve weak equivalence.

For the fibred sequence, take opposites. We have

$$C_u^{\mathrm{op}} = \int_{\mathbb{N}}^{\mathrm{opf}} D^{\mathrm{op}}, \qquad (C_\infty)^{\mathrm{op}} = \mathrm{colim}_n\, D(n)^{\mathrm{op}}.$$

Opposition preserves colimits and the weak homotopy type of a nerve. The preceding argument therefore proves the second assertion. Both constructions and their comparison split as coproducts over the connected components of A. For $A = \varnothing$ both categories are empty and the comparison is the identity. □

A.3. An exact base condition uniform in the fibration PROVED. Let $u : J \to T$ be a functor of small categories. The following conditions are equivalent: every category u/t, $t \in T$, has weakly contractible nerve; and, for every strict $F : T^{\mathrm{op}} \to \mathrm{Cat}$, the projection

$$p_F : \int_J^{\mathrm{fib}} Fu^{\mathrm{op}} \longrightarrow \int_T^{\mathrm{fib}} F$$

belongs to W_∞. In particular, for $J = \mathbb{N}^{\mathrm{op}}$, this homotopy-initiality condition gives the specified isomorphism $[p_F][q]^{-1} : C_\infty \to C$ in Hot. It is sufficient for one particular F that $B(u/t)$ be weakly contractible for every t with $F(t) \neq \varnothing$.

Proof. For $c = (t, z)$ in $\int_T^{\mathrm{fib}} F$, the category p_F/c maps to u/t by forgetting its fibre data. Above $(j, \alpha : u(j) \to t)$ the remaining data are an object of $F(u(j))/F(\alpha)z$. That category has terminal object $\mathrm{id}_{F(\alpha)z}$. There is also an explicit section of the forgetful functor, selecting this terminal object, and a natural transformation from the identity to the section after the projection. Explicitly, the given arrow $x \to F(\alpha)z$ supplies the component from $(j, x, \alpha, x \to F(\alpha)z)$ to the selected object. Thus $B(p_F/c) \simeq B(u/t)$ by actual functors and a natural transformation. Quillen's Theorem A proves sufficiency.

For necessity, fix $t \in T$ and take the discrete-valued presheaf $F_t(s) = \mathrm{Hom}_T(s, t)$. Its fibred construction is T/t, with terminal object (t, id_t), and the pulled-back construction is u/t. The universal hypothesis on p_{F_t} forces $B(u/t)$ to be weakly contractible. This proves the

equivalence without assuming that parallel reindexing functors are homotopic. Necessity is claimed for the universal quantifier over all F, not for a particular fibration or for the restricted subclass satisfying VII's homotopy hypothesis. □

A.4. Preorder cofinality and parallel-map homotopies do not suffice REFUTED. The hypotheses printed at VII ms. 99–101 do not imply that $C_u \to C$ or $C_\infty \to C$ is a weak equivalence. This remains false when T has one object and the fibre category is terminal. Thus VIIb-C026 requires a condition such as A.3, or a proof specific to the path fibration, beyond cofinality in the associated preorder.

Proof. Let $T = \mathbf{B}C_2$, the category with one object and endomorphisms $1, s$, $s^2 = 1$. Take F constant at the terminal category and let $u : \mathbb{N}^{\mathrm{op}} \to T$ send every arrow to 1. The preorder of T is a singleton, the sequence is cofinal in it, and all reindexing functors on the terminal fibre coincide. Nevertheless $C = T$, $C_u = \mathbb{N}^{\mathrm{op}}$, and $C_\infty = e$. The last two nerves are contractible whereas $\pi_1(BT) = C_2$. The multiplication table and normalized chain differentials are independently checked in `code/a_finite_checks.py`; they give $H_1(BT;\mathbb{Z}) = \mathbb{Z}/2$. Grothendieck already identifies the one-object obstruction himself at VII ms. 109. It is not a correction to that page; it refutes the insufficient earlier general hypothesis. □

A.5. The totally aspherical shape does not control colimits in Cat REFUTED. Let $A = \Delta_{\mathrm{inj}}$ have objects $[n] = \{0, \dots, n\}$, $n \geq 0$, and strictly increasing functions. Let $D([n])$ be the one-object category with endomorphism monoid $\mathbb{N}^{n+1}$. An injection sends its standard generators to the indicated standard generators. Then

$$\mathrm{colim}_A^{\mathrm{Cat}} D = \mathbf{B}\mathbb{N}, \qquad H_k\left(B\int_A^{\mathrm{opf}} D;\mathbb{Z}\right) = \mathbb{Z} \quad (k \geq 0).$$

Here $\mathbf{B}\mathbb{N}$ denotes the one-object category; its classifying space has homotopy type S^1. Thus q_D is not a weak equivalence, despite the total W_∞-asphericity of A.

Proof. The injections $[0] \to [n]$ identify all the standard generators in any cocone, and connectedness identifies all the single objects. A cocone into an arbitrary category is therefore exactly an object with one chosen endomorphism. This is the universal property of the one-object free monoid $\mathbb{N}$, proving the claimed category colimit, without using a colimit in commutative monoids instead.

The homomorphism $\mathbb{N}^r \to \mathbb{Z}^r$ induces a weak equivalence on classifying spaces. One direct proof is the covering category with objects $\mathbb{Z}^r$ and a unique arrow $v \to w$ when $w - v \in \mathbb{N}^r$: its underlying poset is filtered, its nerve is contractible, and the free $\mathbb{Z}^r$-action has quotient nerve $B(\mathbf{B}\mathbb{N}^r)$. Consequently the integral homology of $BD([n])$ is $\Lambda^k\mathbb{Z}^{n+1}$. For $k \geq 1$ the ordered wedge basis gives an isomorphism of covariant $\mathbb{Z}A$-modules

$$H_k(BD(-);\mathbb{Z}) \cong \mathbb{Z}[\mathrm{Hom}_A([k-1], -)].$$

These are projective representable modules, whose colimit is $\mathbb{Z}$ and whose higher derived colimits vanish. For $k = 0$ the module is constant $\mathbb{Z}$. The functor $S([n]) = [n+1]$ which adds a new last element admits natural transformations $\mathrm{id}_A \Rightarrow S \Leftarrow \mathrm{const}_{[0]}$, using respectively the original elements and the new last element. Therefore BA is contractible and the higher derived colimits of this degree-zero module also vanish.

Thomason's homotopy-colimit comparison, explicitly restated in Cegarra (2020, Section 3.1, equation (9)), identifies the total nerve with the homotopy colimit. The bar double complex gives the first-quadrant spectral sequence

$$E^2_{p,q} = L_p \operatorname{colim}_A H_q(BD(-);\mathbb{Z}) \implies H_{p+q}\left(B\int_A^{\mathrm{opf}} D;\mathbb{Z}\right).$$

Every column except $p=0$ vanishes and $E^2_{0,q}=\mathbb{Z}$ for every $q\geq 0$, so it has no differential or extension ambiguity. In particular the source of Bq_D has $H_2=\mathbb{Z}$ and its target has $H_2=0$. This verifies the counterexample proposed after TOTASPH. It does not require the full exactness theorem for arbitrary abelian-group-valued diagrams. The accompanying program checks the ordered wedge/representable identification and the colimit relations through [5]; the proof above, rather than extrapolation from those finite checks, establishes the assertion in all degrees.

We verify the total-asphericity premise in A.6 below, recording the argument rather than treating the earlier AI-produced answer as an authority. □

A.6. Verification of the Delta-injective premise PROVED. The category Δ_{inj} has exact colimits of abelian-group-valued diagrams and is totally W_∞-aspherical, but is not filtered. These are the assertions of TOTASPH C.7–C.10 used in A.5.

Proof. We first check the finite algebraic construction which underlies the inherited proof. For every $n\geq 0$ there is an integral combination t of injections $[n]\to[M]$, of augmentation one, such that th is independent of the injection $h:[m]\to[n]$ for each fixed m. Here negative coefficients are allowed. To construct it, for each $k\geq 1$ choose a homogeneous integral polynomial $C_{n,k}(z_0,\dots,z_{n+1})$ having value one at all variables equal to one and such that $y_i^k D_i C_{n,k}$ is independent of i. The map D_i identifies z_i, z_{i+1} and relabels the remaining variables $y_0,\dots,y_n$. Set $C_{0,k}=1$. Inductively put $Q=C_{n-1,2k}$ and prescribe

$$D_i C_{n,k} = Q\prod_{l\neq i} y_l^k.$$

These prescriptions agree on pairwise intersections. For adjacent indices both products agree after the second identification; for nonadjacent indices the required identity is precisely $w_p^{2k}D_pQ = w_q^{2k}D_qQ$. To see that agreement is sufficient over the integers, change coordinates to $x_i=z_i-z_{i+1}$ and $u=z_{n+1}$. For compatible prescribed residues F_i modulo (x_i), let ρ_S set $x_i=0$ for $i\in S$ and set

$$C=\sum_{\varnothing\neq S\subseteq\{0,\dots,n\}} (-1)^{|S|+1}\rho_S F_i \qquad (i\in S).$$

Compatibility makes each term independent of $i\in S$; cancellation modulo (x_j) leaves F_j. Homogeneity is preserved. The degree satisfies $d(n,k)=d(n-1,2k)+nk=k(2^{n+1}-n-2)$.

An injection $g:[n]\to[M]$ corresponds to the monomial with exponents $g(0), g(1)-g(0)-1,\dots,g(n)-g(n-1)-1, M-g(n)$, of total degree $M-n$. Precomposition with the coface omitting i acts on this polynomial by $P\mapsto y_iD_iP$. Choose $P=C_{n,1}$ and $M=n+d(n,1)$. All $t\delta^i$ are equal. The coface identity $\delta^{b+1}\delta^a=\delta^a\delta^b$ then shows that this common value equalizes all injections into $[n-1]$, by downward induction; factoring a proper injection through a coface proves the assertion for every h. The augmentation is the evaluation at all variables equal to one, hence is one.

Let $\mathrm{Aff}_{\mathbb{Z}}(A)$ have the objects of A and augmentation-one finite integral arrow combinations as its arrows. Any two objects of A have a common successor. The combination t just

constructed equalizes any two parallel affine arrows, because both have coefficient sum one. Thus $\mathrm{Aff}_{\mathbb{Z}}(A)$ is filtered. An abelian-group-valued functor on A extends uniquely by linearity to this affine category. Cocones over the two diagrams are the same: respecting ordinary arrows implies respecting affine combinations of augmentation one. The colimits are therefore canonically the same. Since filtered colimits of abelian groups are exact, so are colimits over A.

For a covariant $H : A \to \mathrm{Set}$ and a diagram V of abelian groups on its category of elements, the assignment $(\Sigma_H V)(a) = \bigoplus_{x \in H(a)} V(a, x)$ is exact and has $\mathrm{colim}_A \Sigma_H V = \mathrm{colim}_{\mathrm{El}_A H} V$ by the cocone universal property. Hence colimits on every connected component E of every such element category are exact, and the bar resolution gives $H_j(BE; \mathbb{Z}) = 0$ for $j > 0$. The same argument applies to categories of elements over E: their iterated element construction is an element construction over A. For a cyclic subgroup $L \leq \pi_1(BE)$, the associated covering is the element category of a locally constant set-valued functor on E. It has $H_1 = L_{\mathrm{ab}} = L$ and must be acyclic. All cyclic subgroups are thus trivial. Each BE is simply connected and integrally acyclic, hence weakly contractible by Hurewicz.

If $f : A \to B$ is ordinary final, its comma category $b\backslash A$ is a nonempty connected element category for every b, and therefore is weakly contractible. Quillen's Theorem A makes Bf a weak equivalence. We already proved BA contractible, so BB is contractible. This is total asphericity in the final-image convention of the prompt. Finally, the two arrows $[0] \rightrightarrows [1]$ cannot be equalized by postcomposition with an injection, proving nonfilteredness. □

A.7. Bounds on the class of universally correct Cat-colimit shapes PROVED. Let $\mathcal{C}$ be the class of nonempty connected small categories A for which $q_D \in W_\infty$ for every $D : A \to \mathrm{Cat}$. Then

$$\{\text{filtered categories}\} \subseteq \mathcal{C} \subsetneq \{\text{totally } W_\infty\text{-aspherical categories}\}.$$

The right inclusion is strict by A.5–A.6. For arbitrary, possibly disconnected A, the comparison property is componentwise and the empty shape satisfies it.

Proof. The first inclusion is A.2. For the second, apply the comparison property to every discrete-valued diagram $H : A \to \mathrm{Set}$. Its Grothendieck construction is $\mathrm{El}_A H$ and its colimit is the discrete set $\mathrm{colim}_A H$. Every connected component of the element category must therefore be weakly contractible. In particular the constant singleton diagram makes BA contractible. For an ordinary final $f : A \to B$, use $H_b(a) = \mathrm{Hom}_B(b, f(a))$ and Quillen's Theorem A exactly as at the end of A.6. Then BB is contractible, which proves total asphericity. The componentwise and empty assertions follow from the coproduct decompositions in A.2. □

A.8. Exact classification of Cat-colimit shapes OPEN. We do not determine whether every member of $\mathcal{C}$ in A.7 is filtered. More generally, an intrinsic criterion for membership in $\mathcal{C}$, sharper than the proven bounds, is unresolved in this paper. The finite-object and weakly-terminal scopes reduce to filteredness by the independently checkable total-asphericity criterion TOTASPH C.13; we do not need that additional criterion in the preceding arguments. The first scope not decided by those results consists of infinite categories with no weakly terminal object. No smallest counterexample is asserted.

The simplicial-set criterion cannot prove the missing implication: the map it controls has target $\mathrm{colim}_A ND$, whereas the target needed here is $N \,\mathrm{colim}^{\mathrm{Cat}}_A D$. A.5 explicitly separates these targets. Nor does exactness of abelian-group colimits suffice, since A.6 verifies that property for the same counterexample. The searches and the exact status of the sources are recorded in the report.

A.9. A finite obstruction to replacing Hot by Cat hot PROVED. Let P be the face poset of the boundary of a triangle, with three vertices and three edges, and let $\mathrm{sd}\, P$ be the

poset of nonempty chains in P, ordered by inclusion. The last-element map $\ell : \operatorname{sd} P \to P$ is a weak equivalence and is not invertible in Cat hot. There are exactly 456 monotone maps $P \to \operatorname{sd} P$, and all induce degree zero on the circle.

Proof. The objects of P are the six faces

$$\{0\}, \quad \{1\}, \quad \{2\}, \quad \{0,1\}, \quad \{1,2\}, \quad \{0,2\}.$$

Its arrows are their inclusions. The objects of $\operatorname{sd} P$ are their nonempty chains, with inclusions as arrows. Compositions are the unique arrows given by inclusion. Their nerves are respectively cycles with six and twelve edges. The standard subdivision homeomorphism and the simplexwise linear homotopy to the last vertex show that $B\ell$ has degree one. Every monotone map in the reverse direction gives a simplicial map which sends each of the six source edges to one target edge or a vertex. The total winding has absolute value at most $6/12 < 1$; being integral, its degree is zero. It therefore cannot be a homotopy inverse even after passing to nerves. Each natural transformation induces a homotopy of nerves, and the same holds for any finite zigzag, so no inverse exists in Cat hot. The program `code/a_finite_checks.py` verifies both partial orders, enumerates all vertex images and all compatible edge images, and computes every winding number. Its output is 456 maps of degree zero and degree one for ℓ. This verifies POLES D.6 and provides a finite warning for VIIc-A046 and V-I066; it does not by itself decide the particular path comparison q_L. □

2 The comparison with hammock localization

2.1 The hammock comparison and the scope of the remaining inclusion problem

B.1. The two diagrams and the actual inclusion PROVED. For a type $\tau = \varepsilon_1 \cdots \varepsilon_n$, write I_τ for the free category on vertices $0, \ldots, n$ and the directed edges prescribed by its signs. A type map is a nondecreasing endpoint-preserving functor. Equivalently, its ith edge is sent to a string of $l_i \geq 0$ edges of the same sign; the target word is $\varepsilon_1^{l_1} \cdots \varepsilon_n^{l_n}$. Let Σ be a wide subcategory of a small category M.

Let $F_\tau(a,b)$ be the full category of functors $I_\tau \to M$ with endpoints a, b and backward edges in Σ, with transformations equal to identities at both endpoints. Let $F_\tau^\Sigma(a,b)$ have the same objects and only transformations whose components all lie in Σ. These are strict contravariant diagrams on Ty. Their fibred Grothendieck constructions are respectively $\mathrm{Ch}_\Sigma(M; a, b)$ and $\mathrm{Ch}_\Sigma^\Sigma(M; a, b)$. The inclusion of diagrams induces the specified identity-on-objects functor

$$J : \mathrm{Ch}_\Sigma^\Sigma(M; a, b) \longrightarrow \mathrm{Ch}_\Sigma(M; a, b).$$

Both constructions and J are strictly compatible with concatenation and empty-path units.

Proof. An arrow $(\tau, c) \to (\sigma, d)$ is a pair (u, η) with $u : I_\tau \to I_\sigma$ a type map and $\eta : c \Rightarrow du$ an endpoint-fixed natural transformation. In the restricted category, every component of η must lie in Σ. Its composite with $(v, \theta) : (\sigma, d) \to (\rho, e)$ is $(vu, (\theta u)\eta)$. Composition closure of Σ proves that these arrows form the restricted category. Precomposition sends a backward edge to an identity or to a composite of backward edges, so both diagrams preserve their stated object conditions. It also preserves the vertical restriction.

Concatenation glues the two endpoints literally and concatenates the two type maps and the two component lists. The common component is an identity. The displayed composition formula commutes with this operation. Concatenation of finite ordered lists is strictly associative, and the empty list is a strict unit. This verifies the claims without identifying an equivalence of classifying spaces with an equivalence of categories. □

B.2. A strict and composition-compatible hammock reduction KNOWN. For every small relative category with a wide subcategory Σ, there is a specified simplicial map

$$\rho_{a,b} : N\,\mathrm{Ch}^{\Sigma}_{\Sigma}(M; a, b) \longrightarrow L^{H}(M, \Sigma)(a, b)$$

which is a weak homotopy equivalence. It is natural in relative functors and forms, as a, b vary, a simplicial functor that is the identity on objects and strictly respects concatenation and units. No two-out-of-three or saturation assumption is needed.

Proof. The fixed-shape restricted category is exactly the category of width-zero and width-one hammocks of Dwyer and Kan, Definition 5.1. Their indexing category **II** is $\mathrm{Ty}^{\mathrm{op}}$, with the word-reading convention reversed if one writes composites in the opposite order. Proposition 5.5 identifies the hammock mapping simplicial set with

$$\operatorname*{colim}_{\tau \in \mathrm{Ty}^{\mathrm{op}}} NF^{\Sigma}_{\tau}(a, b), \tag{B.2.1}$$

where the colimit is in simplicial sets, not in categories. Their Proposition 2.2 identifies this localization with simplicial localization (Dwyer and Kan, 1980a).

We specify the map from the manuscript's fibred convention. An n-simplex of its nerve is a string of path morphisms with types $\tau_0 \to \tau_1 \to \cdots \to \tau_n$. Pull all $n+1$ paths back to τ_0 and use the given transformations to form a width-n restricted hammock. Send it to (B.2.1). All face operations except the zeroth simply compose or omit a row. The zeroth face changes the chosen common type from τ_0 to τ_1; the two resulting simplices agree in (B.2.1) by the type-change relation. Degeneracies insert identity rows. Thus this is an actual simplicial map ρ, not merely a map after localization. Pullback to the first type and reduction commute with concatenating two strings of path morphisms, proving the strict composition assertion.

For clarity, we verify the homotopy-colimit issue in this exact comparison. The type category is Reedy with degree the number of edges, direct maps injective on vertices, and inverse maps surjective on vertices. The factorization is through the image on vertices. In $\mathrm{Ty}^{\mathrm{op}}$, latching maps for $\tau \mapsto NF^{\Sigma}_{\tau}$ are the inclusions of the union of the subsets in which one or more specified horizontal edges are identities. Intersections correspond to collapsing the union of those edge sets. Hence these are monomorphisms and the diagram is Reedy cofibrant.

The Reedy category $\mathrm{Ty}^{\mathrm{op}}$ has fibrant constants. Indeed, a proper injective endpoint-preserving map into I_τ has to contain every turning vertex and both endpoints. If τ is not reduced, the inclusion of this reduced subinterval is initial among such maps; if it is reduced, there are no proper such maps. The relevant inverse latching category is therefore empty or connected. Consequently ordinary colimit computes the derived colimit of this diagram. This is the argument of Arakawa and Cnossen, Proposition 4.8 and Lemma 4.16; their Theorem 4.7 gives the simplicial-category comparison, and Definition 1.7, Theorem 1.9, and Corollary 1.11 identify the resulting anima with the mapping space of localization (Arakawa and Cnossen, 2026). Their Remark 4.18 explicitly records the Grothendieck-construction variant.

Finally the natural Grothendieck homotopy-colimit comparison identifies $B\,\mathrm{Ch}^{\Sigma}_{\Sigma}$ with the homotopy colimit of these fibre nerves. The map to (B.2.1) is its canonical colimit cocone, as is also seen from the first-type formula above. The Reedy calculation proves that this specified map is a weak equivalence. For the natural Grothendieck comparison one may use Chachólski and Scherer, Theorem 26.8 and Corollary 26.9 (Chachólski and Scherer, 2001). The Reedy proof and Dwyer–Kan's original definition apply to any wide subcategory; the convention that a relative ∞-category contains all equivalences in the more recent paper is not needed to add an unstated hypothesis here. □

B.3. Two-out-of-three does not make J an isomorphism REFUTED. Even for saturated Σ, J need not be an isomorphism of categories. It already fails for $M = [1] = \{0 \to 1\}$ and Σ consisting of identities.

Proof. Take endpoints $0, 1$, type $(+, +)$, and paths $p = (0, 0, 1)$ and $q = (0, 1, 1)$. There is a full path transformation $p \Rightarrow q$ whose components are $(1_0, 0 \to 1, 1_1)$. Its squares commute. This transformation is not in the restricted category. There are two objects and three arrows in M; the only composites involve identities. The identity class is wide, satisfies two-out-of-three, and is saturated because localization changes nothing and the nonidentity arrow is not invertible. The finite path-category enumerator includes this category and retains the distinct arrows determined by type maps. This example refutes only the claimed categorical isomorphism: B.4 proves that its J is a weak equivalence. □

B.4. Denominators that are already isomorphisms PROVED. Suppose every arrow of Σ is an isomorphism in M; Σ need not contain every isomorphism. Then both path categories have homotopy-discrete nerves, with components canonically $\mathrm{Hom}_M(a, b)$, and J is a weak equivalence. The full all-type category is already isomorphic to this discrete category in $\mathrm{Cat}_{\mathrm{hot}}$. In particular Lead 2 holds for $\Sigma = \mathrm{Iso}(M)$. For $\Sigma = \mathrm{Mor}(M)$, J is the identity and B.2 applies directly.

Proof. Invert the backward arrows in M and compose. This defines a functor $e : \mathrm{Ch}_\Sigma(M; a, b) \to \mathrm{Hom}_M(a, b)_{\mathrm{disc}}$. Naturality of a ladder, with identity endpoint components, implies equality of its two effects. Precomposition by a type map leaves the effect unchanged. Let i insert an arrow as a one-edge forward path. Then $ei = 1$.

There is a uniform contraction, stronger than contractibility after realization. For a path c of type τ, let $\mathrm{pad}(c)$ append a forward identity edge at b, so its type is $\tau * (+)$. On arrows use the old type map concatenated with the identity of $(+)$, and append the identity endpoint component. Collapsing the appended edge gives a natural transformation $\mathrm{pad} \Rightarrow 1$.

Define $T(c)$, also of type $\tau * (+)$, to be a at every old vertex, b at the new last vertex, the identity on every old edge, and $e(c)$ on the appended forward edge. On a path morphism (u, η) use $u * 1_{(+)}$ and identity components; the two effects are equal, so this is a functor. The accumulated prefix effects $a \to c_j$, calculated by inverting the backward arrows, give an endpoint-fixed transformation $T(c) \Rightarrow \mathrm{pad}(c)$. The old last component is $e(c) : a \to b$ and the new last component is 1_b, so the final square also commutes. A path morphism preserves every accumulated prefix after reparametrization; this follows by induction across its commuting squares, using inverse backward arrows. Thus these transformations are natural in c. Collapsing the entire old word to the source of the appended edge gives a natural transformation $T \Rightarrow ie$ with identity components. We obtain the finite zigzag

$$1 \Longleftarrow \mathrm{pad} \Longleftarrow T \Longrightarrow ie.$$

It proves that e and i are inverse in $\mathrm{Cat}_{\mathrm{hot}}$, not merely that Be is a weak equivalence. Empty paths and empty Hom-sets satisfy the same formulas, with the unique functor on an empty category in the latter case.

Inverting arrows that are already equivalences does not change the ∞-category NM. B.2 therefore identifies the restricted nerve with the same discrete mapping space. Its effect map is the same canonical map on components. Since J commutes with the two effect maps, two-out-of-three for weak homotopy equivalences proves the claim. The two-out-of-three principle here concerns maps of spaces, not an additional requirement on Σ.

The parenthetical explanation in Lead 2 cannot mean that the one-edge path is an initial object of the total all-type category: there need not be a type map $(+) \to \tau$. The displayed uniform contraction supplies the required correction and establishes the stronger statement in $\mathrm{Cat}_{\mathrm{hot}}$. □

B.5. What a calculus of fractions identifies directly KNOWN. If (M, Σ) admits the homotopy calculus of two-sided fractions of Dwyer–Kan Definition 6.1, the nerve of the restricted three-arrow category of diagrams

$$a \xleftarrow{\Sigma} x \longrightarrow y \xleftarrow{\Sigma} b$$

computes $L^H(M, \Sigma)(a, b)$. If Σ has two-out-of-three, the full and restricted categories of these three-arrow diagrams are equal. Corresponding statements hold for the two-arrow fraction categories under the left or right calculus, with the variance of Dwyer–Kan's notation retained. These statements alone do not identify the inclusion of the three-arrow category into the full all-type path category.

Proof. The first assertion is exactly Dwyer and Kan, Proposition 6.2(i), with the one-sided versions in (ii)–(iii) (Dwyer and Kan, 1980a). A vertical component at x lies in Σ because its composite with the upper or lower arrow to a is the other arrow to a. The same argument, on the arrow from b, handles the component at y. This proves equality of the two three-arrow categories when two-out-of-three holds. It also proves the one-sided equalities with the appropriate single interior component.

Model categories give the standard homotopy-function-complex interpretation of the hammock side: Dwyer and Kan, *Function complexes*, Propositions 4.4–4.5 and Corollary 4.7, identify it with resolution complexes and, for a cofibrant source and fibrant target in a simplicial model category, the simplicial mapping object (Dwyer and Kan, 1980b). For categories of fibrant objects the fraction statement is Arakawa–Cnossen, Theorem 3.26, which assumes their stated Brown-type axioms and requires no functorial factorization (Arakawa and Cnossen, 2026).

All these comparisons concern the restricted hammock construction or the specified short fraction category. The implication to an arbitrary-vertical all-type category would require proving that its additional inclusion is a weak equivalence. Neither endpoint forcing in the three-arrow shape nor Proposition 6.2 proves that additional assertion. We retain this distinction in B.8. □

B.6. The full category has a canonical map to the localization mapping space PROVED. For $\mathcal{C} = L(M, \Sigma)$ and $a, b \in M$, there is a natural map of spaces

$$Q_{a,b} : B\,\mathrm{Ch}_\Sigma(M; a, b) \longrightarrow \mathrm{Map}_{\mathcal{C}}(a, b)$$

compatible with concatenation. Its composite with BJ is the canonical hammock comparison of B.2. Consequently BJ has a left inverse in Hot and induces injections on homotopy groups at every source basepoint. Both categories have the same component set $\mathrm{Hom}_{M[\Sigma^{-1}]}(a, b)$, and J induces that identity bijection.

Proof. For each type τ, let $p(\tau)$ be the number of forward edges and let $q_\tau : I_\tau \to [p(\tau)]$ count the forward edges preceding a vertex. It is the localization of the interval at its backward edges. This elementary interval localization is also specified in Arakawa–Cnossen, Notation 1.2 (Arakawa and Cnossen, 2026). Precomposition therefore gives an equivalence from $\mathrm{Fun}([p(\tau)], \mathcal{C})$ to the full subcategory of $\mathrm{Fun}(I_\tau, \mathcal{C})$ sending backward edges to equivalences. It commutes with the two endpoint evaluations. On their ∞-categorical fibres at (a, b), composing along $0 \to p(\tau)$ gives a functor to the space $\mathrm{Map}_{\mathcal{C}}(a, b)$.

We make the simultaneous coherence explicit. The ordinals $[p(\tau)]$ form a functor on Ty: a type map sends each forward edge to its same-sign forward string, and the resulting monotone map of ordinals satisfies $q_\sigma u = \overline{u} q_\tau$. The endpoint arrows $[1] \to [p(\tau)]$ commute with all

these maps. Therefore the precomposition equivalences above assemble to an equivalence of diagrams on $\mathrm{Ty}^{\mathrm{op}}$ with values in Cat_∞, after taking the specified endpoint fibres. We invert this *equivalence of diagrams*, rather than choose inverses separately at each type. Inversion with its unit and counit is contractible in the functor ∞-category. The nerve of $F_\tau(a,b)$ maps naturally to the relative-diagram fibre, because its endpoint components are identities. Composing this map of diagrams with the inverse diagram equivalence and the endpoint-arrow composition map gives a coherent map to the constant diagram $\mathrm{Map}_{\mathcal{C}}(a,b)$. Groupoid completion, pointwise, produces a cocone on BF_τ. Taking its colimit and the natural Grothendieck comparison defines Q. The same diagram construction for concatenated ordinals, with any finite number of marked junctions, supplies composition compatibility and its coherences.

On restricted hammocks this construction is exactly inversion of backward arrows and coherent composition, so QBJ is the canonical localization comparison represented by ρ. The universal localization property, with its structural functor from M retained, fixes this comparison; no arbitrary equivalence of mapping spaces is substituted. B.2 shows that QBJ is a weak equivalence. Inverting it in Hot and composing with Q gives a left inverse of BJ. The assertion about homotopy groups follows, with the path from the chosen compatible comparison providing the basepoint identification.

For completeness, the component assertion does not require higher comparisons. The effect of a path is its word in the ordinary localization. Identity insertion, same-direction composition, and cancellation of a marked arrow against its reversed traversal are realized by path morphisms or zigzags of them. A commuting square at a backward arrow uses only inversion of that backward arrow; no vertical component needs to be inverted. These are precisely the relations in ordinary localization. Thus the effect is surjective and identifies exactly the zigzag components of the full category. The restricted assertion is Dwyer–Kan Proposition 3.1. The component identifications commute with J. □

B.7. The elementary grid move and its limited consequence PROVED. A commuting square with backward horizontal arrows in Σ and arbitrary vertical arrows gives, after identity padding, a restricted hammock between the two mixed boundary paths. This does not require two-out-of-three.

Proof. Write the square as

$$w : x' \to x, \qquad w' : y' \to y, \qquad v : x \to y, \qquad v' : x' \to y', \qquad vw = w'v', \quad w, w' \in \Sigma.$$

The two paths $x \xleftarrow{w} x' \xrightarrow{v'} y'$ and $x \xrightarrow{v} y \xleftarrow{w'} y'$ can be padded to type $(-,+,-)$ as

$$x \xleftarrow{w} x' \xrightarrow{v'} y' \xleftarrow{1} y', \qquad x \xleftarrow{1} x \xrightarrow{v} y \xleftarrow{w'} y'.$$

The component list $(1_x, w, w', 1_{y'})$ is a restricted natural transformation from the first padded path to the second. Its middle square is the given equation; its other squares are identity equations. For a forward horizontal square, the two mixed routes compose to the same arrow and are related by type contractions with identity components. Sweeping these local moves through a finite ladder proves its equality on components using restricted paths. A proof of a weak equivalence of entire nerves would additionally have to organize these sweeps on all simplices, compatibly with faces, degeneracies, and compositions. The two-dimensional square calculation itself does not supply that missing assertion. □

B.8. The unrestricted-vertical inclusion beyond the proved cases OPEN. This paper does not settle whether the specified functor J is in W_∞ for every wide Σ, for every Σ satisfying two-out-of-three, or for every saturated Σ. It also does not settle that assertion

for all relative categories admitting a homotopy calculus of fractions, all model categories, or all Brown categories of fibrant objects. The special cases proved in B.4 and the exact short-fraction statements in B.5 retain their stated scope. No counterexample to the general weak-equivalence assertion is established here.

Status and exact remaining step. B.2 identifies the restricted side for every pair. B.6 gives a canonical retraction on homotopy types and the correct components on the full side; it does not prove that the full side has no additional higher homotopy. B.7 removes each arbitrary ladder at the level of one-dimensional homotopy classes. The attempted general argument would require a contractible category of compatible grid sweeps for every higher simplex and compatibility under all simplex operations. That coherent replacement construction has not been proved here. In the model and Brown cases, reducing the restricted mapping space to a short fraction category leaves the same additional comparison into the full all-type category; a theorem about the restricted category cannot be used to close that gap.

A concrete proper marked example of the smallest object size remaining in our tested family, not covered by B.4 and left undecided by these arguments, is the chain $0 < 1 < 2$ with identities and $0 \to 2$ marked. It satisfies two-out-of-three. Its marked arrows are saturated: the functor to the ordinary category generated by a split inclusion $i : A \to B$ and retraction $r : B \to A$ with $ri = 1_A$ and $ir \neq 1_B$, sending $0, 2$ to A, 1 to B, and the two generating arrows to i, r, inverts the marked composite while neither generator becomes invertible. Thus no additional arrow of [2] is forced invertible. This is an undecided instance in the present proof, not a proposed counterexample. Our finite computations in B.9 give no discrepancy for it.

The literature search included Dwyer–Kan's *Calculating simplicial localizations* and *Function complexes in homotopical algebra*, Arakawa–Cnossen's manuscript, and searches for "hammock arbitrary vertical maps" and "Grothendieck path categories simplicial localization". The search located the same question in a 2020 discussion, but no checked primary theorem closing the unrestricted-vertical step. We make no assertion that it is an open problem in the published literature, and no claim of priority. □

B.9. What the finite search establishes PROVED. For the explicitly specified search family and bounds in `code/b_path_search.py`, the full and restricted path-category truncations have equal $H_0(-;\mathbb{F}_2)$ and $H_1(-;\mathbb{F}_2)$. There are 66 relative-category-and-endpoint cases at each of word lengths three and four, for 132 comparisons in total. This finite result proves no unbounded comparison theorem and supplies no counterexample to B.8.

Proof. The program considers the two-element chain, three-element chain, span, cospan, and four-element crown. In each it enumerates all composition-closed wide markings and the two endpoint pairs $(0, 0)$ and $(0, n-1)$, and records whether two-out-of-three holds. For every word of length at most the selected bound it enumerates all relative paths. A morphism retains its full endpoint-preserving type map; two distinct type maps are never identified merely because their source and target paths agree. Poset naturality is tested componentwise, and composites are calculated by composing the vertex maps. Thus all arrows and all composable pairs of the finite truncation are included.

To compute H_1 over $\mathbb{F}_2$, choose a spanning forest of its directed one-skeleton viewed as an undirected multigraph. The remaining edges form generators after the forest is collapsed. Every composable pair contributes the relation consisting of its two edges and its composite, with identities omitted. Gaussian elimination on these binary relations computes the quotient dimension. These are exactly the normalized nerve chain relations in degrees one and two. The code separately records the number of objects, nonidentity arrows, nondegenerate two-simplices, components, and the resulting first Betti number. The two complete output files are

`b_path_search_L3.jsonl` and `b_path_search_L4.jsonl`; all 132 comparisons agree. Higher-length paths can create or kill homology, and the outputs make no persistence claim. □

3 Composition, pointed structures and linear refinements

3.1 Coherent changes of type and the sequential comparisons

The distinction between an ind-isomorphism and a weak equivalence is material here. We use an enrichment of the actual indexing category to retain the coherence that an ind-object in an ordinary homotopy category discards. All path categories in C.P1–C.P4 have arbitrary interior vertical arrows.

C.P1. The enriched category of types PROVED. For types τ, σ, let $P(\tau, \sigma)$ be the category of nondecreasing endpoint-preserving functors $I_\tau \to I_\sigma$, with natural transformations equal to the identity at the endpoints. This is a finite poset. When it is nonempty, its classifying space is contractible. Composition makes these categories the hom-categories of a strict 2-category $\mathcal{T}$. Concatenation extends to a strict monoidal 2-functor on $\mathcal{T}$.

Proof. Thinness follows from thinness of the target interval. We include the contraction from the companion's block at VII ms. 86 because its coherence, and not only its connectedness conclusion, is needed.

Induct on the number of constant-sign runs of σ. For an empty target, the mapping category is terminal. If the first source run has the opposite sign from the first target run, every map is constant on that source run; delete it, retaining its last vertex. Restriction is an isomorphism of mapping categories. We may thus assume that the two initial runs have the same sign ε. Write their vertex intervals as $[0, a]$ and $[0, b]$, with $a, b \geq 1$. The vertex function $h(j) = \max(j, b)$ is an endofunctor of I_σ: it collapses the first run and fixes every subsequent edge.

For $F : I_\tau \to I_\sigma$, put $(RF)(0) = 0$ and $(RF)(i) = hF(i)$ for $i > 0$. A map cannot cross the first target turning vertex within the first source run. Thus RF takes the first source run to $(0, b, \ldots, b)$, is a functor, remains nondecreasing, and preserves endpoints. Applying h to the components away from zero defines R on natural transformations. If $\varepsilon = +$ there is a natural transformation $1 \Rightarrow R$; if $\varepsilon = -$ there is one $R \Rightarrow 1$. All components lie in the first target run or are identities, so their common direction and naturality are valid. The image of R is the full subcategory with prescribed first run $(0, b, \ldots, b)$, and R is the identity on this subcategory. Deleting the prescribed initial runs identifies it with the corresponding mapping category for a target with one fewer run. Induction gives a finite natural-transformation contraction. This proves contractibility without claiming that the chosen contraction is strictly natural under every change of source type.

Vertical and horizontal composition of transformations preserve the endpoint and numerical-order conditions. Gluing at the common endpoint preserves the same conditions and the interchange law. These operations are the stated strict 2-category and strict monoidal structure. □

C.P2. Enriched descent and two final indexing functors PROVED. Let R be the reachability preorder of Ty: its objects are types and it has one arrow $\tau \to \sigma$ exactly when $P(\tau, \sigma)$ is nonempty. For a small relative category (M, Σ) with Σ wide, put

$$F_\tau = \mathrm{Fun}_\Sigma(I_\tau, M)_{a,b}.$$

The diagram $\tau \mapsto BF_\tau$ on $\mathrm{Ty}^{\mathrm{op}}$ admits a coherent extension along the functor $\mathrm{Ty}^{\mathrm{op}} \to R^{\mathrm{op}}$. Moreover, this functor is homotopy final. If $t : \mathbb{N}^{\mathrm{op}} \to \mathrm{Ty}$ is a projective sequence such that every type receives a map from some t_n, its composite $\mathbb{N} \to R^{\mathrm{op}}$ is also homotopy final.

Proof. Precomposition is an actual functor

$$P(\tau,\sigma)\times F_\sigma \longrightarrow F_\tau, \qquad (u,c)\longmapsto cu.$$

For a transformation $\alpha : u \Rightarrow v$ and a path transformation $\eta : c \Rightarrow d$, its component is the composite $cu \xrightarrow{\eta u} du \xrightarrow{d\alpha} dv$. Its endpoint components are identities. Its interior components need not lie in Σ; this is the precise reason for using the full path categories. On objects, precomposition preserves the relative condition because a backward edge is sent to a string of backward edges or to an identity. The functors satisfy the enriched associativity and identity equations strictly.

Apply classifying spaces, using their natural compatibility with finite products. This gives a continuous enriched diagram on the topological category with hom-spaces $BP(\tau,\sigma)$. One completely specified model is to apply the singular functor to these hom-spaces and to BF_τ, and to take the homotopy-coherent nerve of the simplicial category whose hom-spaces are $\operatorname{Sing} BP(\sigma,\tau)$ in the contravariant order. All hom-spaces are Kan complexes. C.P1 makes the map to R^{op} a Dwyer–Kan equivalence: it is the identity on objects and is a weak equivalence on every hom-space, including the empty ones. The coherent nerve therefore gives an equivalence of ∞-categories. Restriction along this equivalence is an equivalence of functor ∞-categories. Consequently the displayed enriched diagram is the pullback of a diagram $\overline{F} : R^{\mathrm{op}} \to \mathcal{S}$, up to a specified coherent equivalence. The space of such lifts with this equivalence retained is contractible. We use precisely these standard facts about coherent nerves and restriction, as stated in Kerodon, §4.6.7, Theorem 4.8.4.1, and Proposition 4.5.4.8 (Lurie, 2026).

We verify finality using actual comma categories, rather than the reachability condition alone. Fix $\sigma \in R$. The comma category for $\mathrm{Ty}^{\mathrm{op}} \to R^{\mathrm{op}}$ at σ is the opposite of the full subcategory

$$\mathrm{Ty}_\sigma = \{\tau \mid \mathrm{Hom}_{\mathrm{Ty}}(\tau,\sigma) \neq \varnothing\}.$$

There is no choice of a map $\tau \to \sigma$ in this comma category: the target is the preorder R. The endofunctor $H(\tau) = \sigma * \tau$ of Ty_σ has natural transformations $H \Rightarrow 1$ and $H \Rightarrow \mathrm{const}_\sigma$, obtained respectively by collapsing the first and second blocks. Their naturality follows directly from endpoint preservation and strict concatenation. They contract the nerve of Ty_σ. Its opposite has the same homotopy type. The comma criterion proves finality.

For the sequence, the comma category at σ is the full subposet of $\mathbb{N}$ consisting of n for which $t_n \to \sigma$ exists. It is nonempty by hypothesis and upward closed because the transition maps are actual type maps. It therefore has a least element and contractible nerve. The same comma criterion proves finality. These are applications of the cofinality theorem with all comma hypotheses checked; see Kerodon, Theorem 7.2.3.1 (Lurie, 2026). □

C.P3. All three comparisons of CATHOT E.26 PROVED. For every small M, every wide $\Sigma \subseteq M$, every $a, b \in M$, and every preorder-cofinal projective type sequence t, let E_t be its fibred Grothendieck construction and $C_\infty(t) = \operatorname{colim}_n F_{t_n}$. The specified functor

$$E_t \longrightarrow \mathrm{Ch}_\Sigma(M; a, b)$$

is in W_∞. The standard map $E_t \to C_\infty(t)$ is also in W_∞. Thus its roof defines a canonical isomorphism

$$\beta_t : BC_\infty(t) \xrightarrow{\sim} B\,\mathrm{Ch}_\Sigma(M; a, b) \quad \text{in Hot}.$$

A strictly commuting map $u : t \to s$ of such projective sequences induces a weak equivalence $u^* : C_\infty(s) \to C_\infty(t)$, and $\beta_t B(u^*) = \beta_s$. The comparisons obtained from $t * s \to t, s$ are independent of comparison choices and transitive in Hot, with their actual maps and roofs understood as above. These assertions settle E.26(1), (2), and (3).

Proof. The natural fibred form of the Grothendieck homotopy-colimit theorem (Chachólski and Scherer, 2001, Corollary 26.9) identifies

$$BE_t \simeq \operatorname*{hocolim}_{n\in\mathbb{N}} BF_{t_n}, \qquad B\,\mathrm{Ch}_\Sigma(M;a,b) \simeq \operatorname*{hocolim}_{\tau\in \mathrm{Ty}^{\mathrm{op}}} BF_\tau.$$

The maps in these comparisons are natural in both the indexing functor and the diagram. To check the variance, the opposite of the fibred construction is the ordinary covariant Grothendieck construction of the opposite fibre categories over the opposite base. The natural realization equivalence between a category and its opposite gives exactly the displayed formula. In particular, the map between these homotopy colimits is the image of the specified functor $E_t \to \mathrm{Ch}_\Sigma$, not an abstractly chosen equivalence.

Apply C.P2. Both homotopy colimits map equivalently to $\mathrm{colim}_{R^{\mathrm{op}}} \overline{F}$. The comparison triangle commutes as a triangle of colimit cocones, with the coherent equivalence of diagrams in C.P2 included. Two-out-of-three proves that the specified map $E_t \to \mathrm{Ch}_\Sigma$ is a weak equivalence. The other map $E_t \to C_\infty(t)$ is a weak equivalence because nerves commute with filtered colimits of categories, and a sequential homotopy colimit of simplicial sets maps weakly equivalently to its ordinary filtered colimit. These arguments do not require fully faithful transition functors.

A natural transformation $u : t \to s$ produces the stated map of sequential colimits. The cartesian arrow

$$(t_n, F(u_n)c) \longrightarrow (s_n, c)$$

in the all-type category gives a natural transformation between $E_s \to E_t \to \mathrm{Ch}_\Sigma$ and the direct inclusion $E_s \to \mathrm{Ch}_\Sigma$. Strict commutation of the sequence maps is exactly its naturality in n. Naturality of the colimit cocones now gives $\beta_t B(u^*) = \beta_s$. Since both β's are isomorphisms, $B(u^*)$ is one. The weak-equivalence assertion about the actual functor follows from the definition of W_∞.

Concatenating two projective sequences gives a projective sequence, with canonical block-collapse projections to both factors. It is preorder-cofinal, since a map from either factor to a type can be preceded by the corresponding block-collapse projection. Hence the resulting comparison is precisely $\beta_t^{-1}\beta_s$. All choices, after any required cofinal reindexing, give this same expression. Transitivity follows by cancelling the middle β. This proof retains the whole enriched action, so the known failure of an ind-isomorphism to imply an equivalence of unions does not apply. □

C.P4. CATHOT E.27 for every admissible class and every cofinal block PROVED. Let τ_0 be a nonempty word containing both signs. Use the manuscript's left-padded, right-padded, and symmetrically padded systems of powers of τ_0, with colimits $L_{\tau_0,\Sigma}$, $R_{\tau_0,\Sigma}$, and $P_{\tau_0,\Sigma}$. The actual maps q_L, q_R from the first two colimits to the third are weak equivalences. On fixed endpoint fibres, the operation

$$\mu_{\tau_0} = B\kappa_{\tau_0}\big((Bq_L)^{-1} \times (Bq_R)^{-1}\big)$$

is associative and unital in Hot. It is independent of τ_0 under the canonical comparisons of C.P3 and agrees with the strict all-type concatenation. No comparison with simplicial localization is needed for this conclusion.

Proof. The numbers of alternating runs of τ_0^n tend to infinity. Every finite sign word is obtained from a sufficiently long power by collapsing unused edges and composing or expanding retained same-sign runs. Thus all three projective sequences satisfy C.P3. The constant-padding maps arise contravariantly from strictly commuting maps of these sequences: for q_L, collapse the

rightmost n blocks of τ_0^{2n}, and for q_R, collapse the leftmost n blocks. They commute with the specified outer-collapse transition maps block by block. C.P3 makes q_L, q_R weak equivalences.

Write $A_{a,b} = B\,\mathrm{Ch}_\Sigma(M; a, b)$ and let $\gamma_{a,b,c} : A_{a,b} \times A_{b,c} \to A_{a,c}$ be strict concatenation after realization, with factors written in traversal order: $\gamma(c, d)$ first traverses c and then d. Thus, if enriched composition is written with factors $A_{b,c} \times A_{a,b}$ and denoted by $*$, the present $\gamma(c, d)$ is $d * c$. The comparisons of C.P3 satisfy

$$\beta_P B q_L = \beta_L, \qquad \beta_P B q_R = \beta_R, \qquad \beta_P B \kappa_{\tau_0} = \gamma(\beta_L \times \beta_R).$$

For the last identity, evaluate on the common-stage diagram with two words τ_0^n. Both maps send the pair to its literal concatenation, of type τ_0^{2n}. Increasing n pads the two outer ends, so the maps commute with the stated transitions. The diagonal $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$ is final, and finite products of spaces commute with filtered colimits. Consequently this common-stage verification identifies the actual maps on the two sequential homotopy colimits. The natural Grothendieck comparison then gives the displayed equality in Hot.

Substituting the first two identities into the third gives

$$\beta_P \mu_{\tau_0} = \gamma(\beta_P \times \beta_P).$$

Strict associativity and the empty-path units for γ imply associativity and the corresponding unit identities for μ_{τ_0}. Every other cofinal block gives the same transported operation on $A_{a,b}$. The construction uses all the coherent comparison maps of C.P2–C.P3, rather than an inference from equality on component-valued finite tests. Words containing only one sign have bounded useful length and are outside the statement; they are not the cofinal blocks used at VII ms. 125–127. □

3.2 A finite obstruction in Cat hot

C.P5. The one-sided comparison need not be invertible modulo natural transformations REFUTED. There is a four-object category X, with $\Sigma = \mathrm{Mor}(X)$ and fixed endpoints $a = b$, for which the actual constant-tail extension $q_L : \mathcal{L}(X; a, a) \to \mathcal{P}(X; a, a)$ of CATHOT E.23 is a weak equivalence but is not invertible in Cat hot. Thus the sequential comparisons cannot in general be promoted from Hot to Cat hot, even for finite X and the canonical two-letter block.

Proof. Take X to be the poset on $\mathbb{Z}/4$ whose only nonidentity arrows are

$$0 \to 1, \quad 0 \to 3, \quad 2 \to 1, \quad 2 \to 3.$$

There are four objects and eight arrows; compositions involving a nonidentity arrow have an identity as the other factor. Its nerve is the four-cycle, so $BX \simeq S^1$. Let $\widetilde{X}$ be the integer fence with arrows from an even integer to its two adjacent odd integers. Reduction modulo four is a covering functor $\widetilde{X} \to X$: every incoming and outgoing base arrow has a unique lift at a chosen vertex. Its nerve is the universal covering line.

Use the same integer fence as the parameter category J. Objects of $\mathcal{P}(X; 0, 0)$ are functors $J \to X$ eventually constant at zero on both tails; arrows are endpoint-fixed natural transformations. Every such path lifts uniquely to $\widetilde{X}$ with left tail zero. A natural transformation lifts uniquely with the same left-tail component, by the arrow-lifting property and its commuting squares. Its right-tail component is an identity, so the two lifted right endpoints agree. The lifted right endpoint is therefore constant on each zigzag component. Denote by $\mathcal{P}_0$ the full subcategory on paths whose right endpoint lift is zero. This definition, rather than a chosen identification of component sets with $\mathbb{Z}$, is all that is needed.

Suppose a functor $r : \mathcal{P}(X;0,0) \to \mathcal{L}(X;0,0)$ were an inverse to q_L in $\mathrm{Cat}_{\mathrm{hot}}$. In particular, there would be a finite zigzag of natural transformations, of some length k, between $1_{\mathcal{P}}$ and $q_L r$. Evaluating this zigzag at a path $c \in \mathcal{P}_0$ produces a zigzag of k endpoint-fixed path transformations. Every intermediate path also has right endpoint lift zero, since each transformation preserves that endpoint. Lift this objectwise zigzag to $\widetilde{X}$ and then evaluate it at any integer $i \geq 0$. The final path $q_L r(c)$ is constant at zero on the nonnegative half-line; because its right endpoint lift is zero, its lift there is also zero. Each arrow of $\widetilde{X}$, in either direction of a zigzag, changes the integer vertex by at most one. Hence the lifted value of c at every $i \geq 0$ would have distance at most k from zero.

For $N \geq 1$, define a path in $\widetilde{X}$ by

$$\widetilde{c}_N(i) = \begin{cases} 0, & i \leq 0, \\ i, & 0 \leq i \leq 4N, \\ 8N - i, & 4N \leq i \leq 8N, \\ 0, & i \geq 8N. \end{cases}$$

At every generating edge of J this is an arrow of the integer fence or an identity. Its reduction modulo four is an object $c_N \in \mathcal{P}_0$, and its lifted value at the positive coordinate $4N$ is $4N$. Choosing $4N > k$ contradicts the preceding bound. No functor r can therefore give the required one-sided inverse in $\mathrm{Cat}_{\mathrm{hot}}$.

The weak equivalence of q_L is C.P4 (and is also the independently checked CATHOT E.25). The program `code/b_crown_checks.py` checks the full finite base category, the covering rule at all residues, all endpoint-fixed transformations of parameter length six, and the explicit path family through $N = 8$. The distance argument proves the obstruction for arbitrary N and arbitrary candidate inverse functors; finite enumeration is not used as a substitute for that quantifier. □

3.3 Structures which do not require a comparison theorem

C.1. Strict composition already gives a coherent enriched category PROVED. For a small relative category (M, Σ), the collection

$$\mathcal{G}(a,b) = N\,\mathrm{Ch}_{\Sigma}(M;a,b), \qquad a,b \in \mathrm{Ob}\,M,$$

has strictly associative and strictly unital simplicial composition. The unit is the empty path. The same is true with Σ-valued vertical components. Applying Ex^{∞} to each mapping simplicial set preserves these equations and makes each mapping object Kan. Consequently each construction determines a category enriched in spaces, and hence a Segal presentation. This statement does not identify the enrichment with $L(M, \Sigma)$.

Proof. For $a, b, c \in M$, concatenation of words, interval diagrams and natural transformations defines a functor

$$* : \mathrm{Ch}_{\Sigma}(M;b,c) \times \mathrm{Ch}_{\Sigma}(M;a,b) \longrightarrow \mathrm{Ch}_{\Sigma}(M;a,c),$$

where the first factor is composed after the second. The natural transformations agree at the glued vertex because both components there are 1_b. The concatenated type map is again endpoint preserving and nondecreasing. The only new identification is the common vertex; no new backward arrow is created. Concatenation of finite words is associative as an equality, including the reindexing convention in Part A. The empty word changes neither the diagram nor any of these maps. Closure of Σ under composition suffices when type maps lengthen a run. The weak-vertical condition is preserved because every component of the concatenation was a component of an input.

The nerve preserves products as an equality of simplicial sets. The functor Ex preserves products as a right adjoint, and filtered colimits of sets commute with finite products. Therefore Ex^∞ preserves finite products and the terminal simplicial set. Apply it to the displayed multiplication and its commuting associativity and unit diagrams. Its usual natural weak equivalence from the identity and its Kan property give the asserted enrichment. For completeness, its Segal nerve in degree n is

$$\coprod_{a_0,\dots,a_n} \prod_{j=1}^{n} \mathrm{Ex}^\infty \mathcal{G}(a_{j-1}, a_j).$$

The Segal maps are isomorphisms. A Reedy fibrant replacement gives a Segal space, since the object space is discrete and the displayed products already compute the relevant homotopy fibre products. Rezk completion changes the presentation, not its associated enriched homotopy theory. These observations establish coherence for the raw model; a comparison must additionally identify its mapping spaces and its action on objects. □

C.2. The hammock enrichment has the localization interpretation KNOWN. For every small relative category (M, Σ), the simplicial category $L^H(M, \Sigma)$ and its mappingwise Ex^∞ replacement have strict associative unital composition, component category $M[\Sigma^{-1}]$, and the localization mapping spaces. Composition with a specified arrow of Σ induces weak equivalences on all relevant mapping spaces. These assertions hold without two-out-of-three or saturation.

Verification of the source and its scope. The construction and composition are Definition 2.1 of Dwyer and Kan (1980a); Proposition 3.1 identifies components and Proposition 3.3 treats composition with a weak equivalence. Part B specifies the Grothendieck-construction version of the weak-vertical model and its comparison. The product-preserving Kan replacement is justified in C.1. None of these theorems asserts that an inclusion permitting arbitrary vertical components is a weak equivalence. Nor does an equivalence of individual mapping spaces, without compatibility, identify a prescribed multiplication. The all-arrows compatibility supplied in CATHOT E.24 and the companion's VII ms. 169 block is used with that precise restriction. □

3.4 Pointed composition and Quillen's groupoid

C.3. The unit for pointed composition is S^0 PROVED. Composition of pointed mapping spaces, with zero morphisms as basepoints, has the form

$$K(b,c) \wedge K(a,b) \longrightarrow K(a,c), \qquad S^0 \longrightarrow K(a,a).$$

The nonbasepoint of the unit is 1_a. Replacing smash product by the Cartesian monoidal structure on pointed spaces cannot express unital composition with absorbing zero whenever $1_a \neq 0$ in $\pi_0 K(a,a)$.

Proof. In the Cartesian monoidal category of pointed spaces the unit is the one-point space, and its only pointed map to $K(a,a)$ selects zero. The left unit axiom would then say $0 \circ f = f$, while absorption says $0 \circ f = 0$. Applying this to $f = 1_a$ contradicts the hypothesis on components. For smash product the unit is S^0, and a based map from it can select the identity at its other point. A map on the product which is constantly zero on either axis descends to the smash product. The associative and unit diagrams descend as well. This is the correction made in IX ms. 17; it is a structural requirement, not a choice of notation. □

C.4. A sufficient pointed model with strict absorption PROVED. Let M be a pointed simplicial model category in a universe in which the objects and replacements under consideration exist. On its full simplicial subcategory of cofibrant-fibrant objects, the simplicial mapping spaces are pointed Kan complexes, their composition factors through smash product, and their units and associativity are strict. They present the pointed derived mapping spaces of M.

Proof. Let 0 be the zero object. Tensoring preserves the initial object, and mapping into the terminal object is unique, so the enriched spaces $\mathrm{Map}_M(0,b)$ and $\mathrm{Map}_M(a,0)$ are each the one-point simplicial set. Their composite selects a zero simplex in $\mathrm{Map}_M(a,b)$ and all its degeneracies. Enriched associativity implies that composing with it in either variable gives that zero map exactly. Thus composition descends levelwise through the quotient by the two axes, which is the smash product of pointed simplicial sets. The identity maps define units from S^0. For cofibrant a and fibrant b, the simplicial model axiom gives the Kan property. The derived identification is Dwyer and Kan (1980b, Corollary 4.7); it is compatible with composition on the cofibrant-fibrant simplicial category, as in their Proposition 4.8. Choosing one cofibrant-fibrant representative for each desired object gives the stated model. This construction fixes a pointed enrichment, rather than choosing unrelated points in spaces already identified up to unpointed equivalence. □

C.5. The precise groupoid comparison with Quillen KNOWN. In a model category, with cofibrant source a and fibrant target b, Quillen's groupoid of maps and homotopy classes of homotopies is naturally equivalent to the fundamental groupoid of a derived function complex. In a simplicial model category this is the fundamental groupoid of $\mathrm{Map}_M(a,b)$. In particular, for actual maps $f, g : a \to b$,

$$\pi_1^{\mathrm{Quillen}}(a,b;f,g) \cong \mathrm{Hom}_{\Pi_1 \mathrm{Map}_M(a,b)}(f,g).$$

In the pointed case the group at the zero map is $\pi_1(\mathrm{Map}_M(a,b),0)$. For arbitrary source and target the formula is applied after replacements, with specified transported maps.

Source check and transport. Quillen (1967) constructs the groupoid in Chapter I, §2, Proposition 1. In the attached transcription this is Proposition 1.2.1. Chapter II, §2, the remark following Proposition 5 explicitly identifies it with the fundamental groupoid of the simplicial mapping complex when the source is cofibrant and the target fibrant; the transcription numbers these as Proposition 2.2.5 and its following remark. For a general model category, the resolution description of function complexes in Dwyer and Kan (1980b, Proposition 4.4) realizes the same cylinder and path homotopies, and their relative homotopies. Its fundamental groupoid is consequently the same one.

A weak equivalence of mapping spaces induces an equivalence of fundamental groupoids, not equality of their chosen objects. To obtain the displayed bijection with the specified f, g, use their vertices under the comparison, or supply paths from their images to the selected vertices. Replacing a chosen basepoint path changes an isomorphism of fundamental groups by the corresponding conjugation. The fundamental groupoid is the formulation free of that ambiguity. Every component of the hammock complex already has zigzag vertices; its fundamental groupoid therefore extends the construction to derived morphisms. For a literal Grothendieck path vertex, the extension requires the comparison in Part B and its agreement with that vertex. This qualification is the remaining model-specific issue in IX-040, IX-041 and IX-072. □

C.6. Loop and suspension identities with their actual hypothesis PROVED. Let $\mathcal{C}$ be a pointed ∞-category. If the pullback $\Omega b = 0 \times_b 0$ exists, then naturally in a,

$$\mathrm{Map}_{\mathcal{C}}(a, \Omega b) \simeq \Omega \,\mathrm{Map}_{\mathcal{C}}(a, b).$$

If the pushout $\Sigma a = 0 \amalg_a 0$ exists, then naturally in b,

$$\mathrm{Map}_{\mathcal{C}}(\Sigma a, b) \simeq \Omega \,\mathrm{Map}_{\mathcal{C}}(a, b).$$

These are pointed equivalences with the zero maps as basepoints. When both constructions exist for every object, they give $\Sigma \dashv \Omega$. The corresponding construction in a pointed derivator gives an adjunction of derivator morphisms; Ω preserves homotopy limits and Σ preserves homotopy colimits in its domain of shapes.

Proof. The defining mapping-space property of the first pullback is

$$\mathrm{Map}(a, \Omega b) \simeq \mathrm{Map}(a, 0) \times^{h}_{\mathrm{Map}(a,b)} \mathrm{Map}(a, 0).$$

Both outside spaces are contractible, and their maps select zero. The right side is precisely the based loop space. The equivalence carries the zero map to its constant loop, since it is induced by the zero cone. The pushout calculation is the dual calculation in the first variable. Combining the two natural equivalences gives the adjunction, including its coherence. On components it yields

$$\mathrm{Hom}_{\mathrm{Ho}\mathcal{C}}(\Sigma a, b) \cong \mathrm{Hom}_{\mathrm{Ho}\mathcal{C}}(a, \Omega b) \cong \pi_1(\mathrm{Map}_{\mathcal{C}}(a, b), 0),$$

with the group operations induced by concatenation of loops.

For a pointed derivator, extension by zero to a corner, followed by the appropriate homotopy Kan extension to a square and evaluation at its opposite corner, constructs these functors simultaneously on all diagram categories. Their adjunction is Groth (2013, Definition 3.16 and Proposition 3.17). The adjoint-morphism criterion and its Kan-extension compatibility in that paper, Proposition 2.9 and its dual, show that the right adjoint preserves homotopy limits and the left adjoint preserves homotopy colimits. Thus IX-084 and IX-087 hold in this setting. Merely writing $A \mapsto M^A[W_A^{-1}]$ has not established that setting. □

C.7. Representing the groups π_1 does not give the loop-space identity REFUTED. Even for a pointed ∞-category, representability of all functors $\pi_1 \,\mathrm{Map}(a, -)$ and $\pi_1 \,\mathrm{Map}(-, b)$ on its homotopy category does not imply the space-level identities of C.6.

Proof. Take the full subcategory of pointed spaces on $*, S^0, K(\mathbb{Z}, 2)$. It is pointed. All of its mapping spaces have trivial fundamental group at zero. Indeed the only relevant nondiscrete one between distinct nonzero objects is

$$\mathrm{Map}_*(S^0, K(\mathbb{Z}, 2)) \simeq K(\mathbb{Z}, 2).$$

Maps from the connected space $K(\mathbb{Z}, 2)$ to the discrete pointed S^0 form a point. The higher homotopy groups of the self-mapping space of $K(\mathbb{Z}, 2)$ are its reduced cohomology groups in degrees $2 - i$; these vanish for $i \geq 1$. The self-mapping space of S^0 is discrete. Maps involving $*$ are contractible. Consequently the two π_1 functors are the constant trivial-group functors and are represented by $*$; one may take both proposed functors on the homotopy category to be the constant-zero functor.

For $a = S^0$ and $b = K(\mathbb{Z}, 2)$ the proposed identity would identify the contractible space $\mathrm{Map}_*(a, *)$ with $K(\mathbb{Z}, 1)$. This is false. The example refutes the inference from the representability of groups, which is exactly what precedes IX ms. 23. It does not claim that a true homotopy pullback has the wrong universal property. The missing datum is the pullback or an equivalent coherent representing property. □

3.5 The linear statement with a specified enhancement

C.8. The requested linear objects for complexes PROVED. Let k be a commutative ring and let $M = \mathrm{Ch}(k)$, with quasi-isomorphisms and cohomological grading. For complexes a, b set

$$K_k(a,b) = \tau_{\le 0}\mathbf{R}\mathrm{Hom}_k(a,b) \quad \in D^{\le 0}(k).$$

There are associative unital composition maps

$$K_k(b,c) \otimes_k^{\mathbf{L}} K_k(a,b) \longrightarrow K_k(a,c), \qquad k \longrightarrow K_k(a,a).$$

Under Dold–Kan, the underlying pointed space is the derived mapping space of complexes, and

$$\pi_i \mathrm{Map}_{L(M,W)}(a,b) = H^{-i}\mathbf{R}\mathrm{Hom}_k(a,b) = \mathrm{Hom}_{D(k)}(a, b[-i]) \quad (i \ge 0),$$

where $i = 0$ denotes the underlying set of the indicated abelian group. The same assertions hold in any specified dg model whose derived mapping complexes and composition are given by such an enrichment.

Proof. Choose a dg-projective replacement $Qa \to a$ for every object, in the working universe. Such replacements and the projective model structure are verified in Keller (2006, Proposition 3.1 and Theorem 3.2). The complexes $\mathrm{Hom}_k(Qa, Qb)$ compute derived Hom and have strictly associative dg composition. For a cochain complex C, good truncation has C^n for $n < 0$, $\ker(d : C^0 \to C^1)$ in degree zero, and zero in positive degrees. Composition restricts to these truncations: negative degrees add to a nonpositive degree and the product of degree-zero cycles is a cycle. The Leibniz identity verifies that the restriction is a chain map, and each 1_{Qa} is a degree-zero cycle.

For the derived tensor assertion we must not assume that these Hom complexes are K-flat. In the closed monoidal derived category, evaluation and its adjunction give

$$\mathbf{R}\mathrm{Hom}(b,c) \otimes_k^{\mathbf{L}} \mathbf{R}\mathrm{Hom}(a,b) \longrightarrow \mathbf{R}\mathrm{Hom}(a,c).$$

The inclusion $D^{\le 0}(k) \to D(k)$ is monoidal: derived tensor of two nonpositive objects is nonpositive. Its right adjoint $\tau_{\le 0}$ is therefore lax monoidal. Apply that structure and the displayed evaluation composition. This constructs the required maps with their associative and unit coherence. Equivalently, the map is the unique factorization through the good truncation of the composition after the two truncation inclusions; its source is nonpositive. Repeating this for any number of factors identifies both bracketings with the same iterated evaluation map.

Reindex the good truncation as a nonnegative chain complex by $C_i = C^{-i}$, and apply the inverse Dold–Kan functor of The Stacks Project Authors (2026, Theorem 14.24.3). Its homotopy groups are $H_i(C) = H^{-i}(C)$. The dg function complex, or a simplicial resolution of the complexes, identifies this simplicial module with the derived function complex. That is the enrichment version of the resolution comparison in Dwyer and Kan (1980b, Proposition 4.4). The zero chain map supplies the basepoint; the dg composition supplies the bilinear pairing. Reversing the order of the displayed tensor factors requires the Koszul symmetry, not an unsigned exchange of homogeneous maps. This proves the claimed refinement of IX-061, IX-065 and IX-070 in the stated dg setting.

For example, $a = k[0]$ and $b = k[1]$ give $K_k(a,b) = k[1]$, concentrated in cohomological degree -1, with mapping space $K(k,1)$. Thus the higher information is present even though the group of degree-zero derived morphisms is zero. In this model $\Sigma a = a[1]$ and $\Omega b = b[-1]$ agree with C.6. □

C.9. The manuscript's unaxiomatized additive generality OPEN. For an arbitrary additive k-linear category M and a wide class Σ closed under finite direct sums, we do not establish that the literal categories $\mathrm{Ch}_\Sigma(M; a, b)$ admit composition-compatible refinements in $D^{\leq 0}(k)$ agreeing with their nerves. Adding two-out-of-three, saturation, or an abstract suspension–loop equivalence on the homotopy category is not proved here to suffice. The exact remaining question is existence of the linear enhancement together with a comparison to those specified path categories.

Scope and failed reduction. At IX ms. 18 Grothendieck explicitly asks for additional compatibility of Σ with addition, and at ms. 19 suggests an equivalence between suspension and loops. These sentences do not specify an axiom system. C.8 proves a precise instance, and it must not replace the unspecified class by all dg model categories without recording the restriction. An additive structure on a homotopy category alone does not construct chain-valued composition or a comparison of its underlying spaces. A stronger established criterion is Dugger and Shipley (2007, Theorem 1.3 and Definition 6.2): a stable, additive, combinatorial model category has enrichment in spectra of simplicial abelian groups. Their definition includes compatible additive cosimplicial resolutions (Definition 6.1), not merely additive Hom sets. Their chain-enrichment results are in §9. This supplies another sufficient enhanced context; it does not verify that an arbitrary relative additive category satisfies those hypotheses. We searched the literature on additive simplicial localization, dg localization, and additive model enrichments. No argument closing the general comparison gap is claimed. □

C.10. Literal pointed and ind-pointed transfer OPEN. For a general pointed relative category, the claims that the manuscript's specified stabilized-path pairings define an associative unital smash enrichment, and that its stage system defines the proposed pointed objects of $\mathrm{Ind}(\mathrm{Hot}_*)$ and $\mathrm{Ind}(\mathrm{Cat}\,\mathrm{hot}_*)$, remain conditional on a pointed, composition-compatible comparison or a direct construction. An unpointed mapping-space equivalence alone does not settle IX-044, IX-049 or IX-052 at this literal level.

The exact outstanding coherence. A chosen zero-arrow vertex gives a pointed nerve, but concatenating it with a nonempty path generally gives a different path object. To factor through smash one needs absorption homotopies in both variables which agree at the two-axis intersection; to assert associativity one needs their compatibility with threefold concatenation and units. For an ind-object one additionally needs these choices compatible with every transition map. C.4 supplies all of this in a strict pointed simplicial model, and C.6 supplies the resulting loop identities. Transferring it to the exact categorical system requires specified comparison maps with these coherences. A bijection on components, or an isomorphism of unpointed ind-objects, provides none of them. The elementary cases $\Sigma = \mathrm{Iso}(M)$ and a chosen dg model have the evident discrete or enhanced pointed refinement; the general literal statement remains on the register. □

4 Rectification of diagrams

4.1 The tests in Chapter IX and a correction to Lead 3

D.1. The typed forms of the three tests PROVED. Let (M, W) be a relative category, let $\mathcal{C} = L(M, W)$, and let $\mathscr{A}$ be a specified class of categories small relative to the universe in which M is being used. Write

$$H_{a,b}(A) = \mathrm{Hom}_{M^A[W_A^{-1}]}(a_A, b_A), \qquad R_A : M^A[W_A^{-1}] \longrightarrow \mathrm{Ho}\,\mathrm{Fun}(NA, \mathcal{C}).$$

Here diagrams are covariant; replace A by A^{op} for the manuscript's coefficient convention. Rectifiability at A means that this specified functor R_A is an equivalence.

At IX ms. 22–23, test (a) asks for representability of the pointed group-valued functor denoted $\pi_1(a,b)$ in one variable, or the other, or both: $\mathrm{Hom}(\Sigma a, b) \cong \pi_1(a,b)$ and $\mathrm{Hom}(a, \Omega b) \cong \pi_1(a,b)$ in $M[W^{-1}]$. These three versions must be distinguished. The second formulation of test (b), on the bounded class $\mathscr{A}$, asks that

$$u^* : H_{a,b}(B) \longrightarrow H_{a,b}(A)$$

be a bijection for every $u : A \to B$ in W_∞ with $A, B \in \mathscr{A}$, and every a, b. The first formulation says that the sequential path categories are "homotopes à des catégories de Thomason-Kan". If "homotopes" means isomorphic in Hot and "Thomason-Kan" means Thomason fibrant, this is true of every small category. It therefore is not a criterion. An isomorphism in Cat hot to an object local for the indicated tests is a different assertion; fibrancy alone cannot replace that localness.

Test (c) asks for the two natural pointed equivalences

$$\mathcal{H}(a, \Omega b) \simeq \Omega \mathcal{H}(a,b), \qquad \mathcal{H}(\Sigma a, b) \simeq \Omega \mathcal{H}(a,b), \tag{D.1.1}$$

separately when the corresponding representing object exists, with compatibility with the loop group object in $\mathrm{Hot}_\bullet$. Here $\mathcal{H}$ must be a specified pointed mapping object; using the manuscript's path object retains the comparison hypotheses of Parts B–C.

Proof. These are the displayed formulas and their quantifiers at IX ms. 22–23. The phrase "voire même des deux à la fois" prevents us from silently replacing (a) by simultaneous existence. The source itself separates the two formulas in (c). The localization functor $NM \to \mathcal{C}$ sends objectwise W-maps of diagrams to equivalences, so its composite with the homotopy-category functor uniquely factors through R_A; this defines its source, target, and variance. Finally, a fibrant replacement $K \to RK$ in the Thomason model structure is in W_∞ and has fibrant target. This proves the assertion concerning the weak reading of the first (b), without asserting a natural-transformation inverse to $K \to RK$. The bounded formulation avoids using a single small category as an unrestricted representative for tests of every size. □

D.2. The finite example does not fail rectifiability REFUTED. The claim in Lead 3 that the finite example at IX ms. 23 fails rectifiability is false. It is rectifiable at every small category, satisfies (a), both meaningful discrete versions of (b), and (c), and its naive family satisfies Kaledin's enhanced-category axioms.

Proof. Let E have objects $0, x$ and arrows

$$1_0, \quad u : 0 \to x, \quad v : x \to 0, \quad 1_x, \quad z = uv : x \to x, \qquad vu = 1_0, \quad z^2 = z \neq 1_x.$$

All remaining composites are determined by identities and these equations; equivalently this is the full category of pointed sets on the one- and two-element pointed sets. Set $W = \{1_0, 1_x\}$. It contains all isomorphisms, is saturated, and satisfies two out of three. The program `d_finite_checks.py` checks all 34 composable triples, the entire composition table, and the zero-object property.

For ordinary categories A, E, in simplicial degree n there is an equality

$$\mathrm{Fun}(NA, NE)_n = \mathrm{Hom}_{\mathrm{sSet}}(N(A \times [n]), NE) = \mathrm{Fun}(A \times [n], E) = N\,\mathrm{Fun}(A,E)_n.$$

These equalities respect faces and degeneracies. Localization at identities changes neither E nor E^A. Consequently R_A is the canonical isomorphism for every A, including the empty

category and categories with nonidentity endomorphisms. This proves rectifiability, not merely agreement on components.

For constant diagrams, $H_{a,b}(A) = \mathrm{Hom}_E(a,b)^{\pi_0 A}$: naturality forces the chosen arrow to be constant along every arrow, hence along each zigzag component. This proves the second (b). The identity-backward path calculation of Lead 2 gives a categorical homotopy equivalence with the discrete Hom set, so the first (b) also holds with a discrete, hence local and Thomason-fibrant, target. All based π_1 groups are trivial. The assignments $\Sigma a = 0 = \Omega a$ represent them, and both sides of (D.1.1) are contractible. The surviving distinct arrows $1_x, z$ show that the localization is not equivalent to a point.

Finally the naive family is the represented family $A \mapsto \mathrm{Fun}(A, E)$. Kaledin's Example 7.2.1.16 applies to every ordinary category E, with no finite-limit hypothesis, and makes its contravariant version an enhanced category. Thus there is no failed Kaledin axiom to name in this example. What it lacks is finite products, not rectifiability or enhancement. □

D.3. The actual finite-product obstruction PROVED. A category with finitely many arrows and all finite products has at most one arrow between any two objects. If it has a zero object as well, it is equivalent to the terminal category.

Proof. Two distinct arrows $a \rightrightarrows b$ would give 2^n distinct arrows $a \to b^n$ for every n, contradicting the fixed finite bound on all arrows. In a pointed category a zero map exists between every pair, so every Hom set is a singleton and every two objects are isomorphic. The proof uses products in the category in question; one cannot infer them from representability of the trivial π_1-functor in D.2. □

D.4. Rectifiability implies the second test (b) PROVED. If R_A is an equivalence for every $A \in \mathscr{A}$, then there are natural bijections

$$H_{a,b}(A) \cong [BA, \mathrm{Map}_{\mathcal{C}}(a,b)]. \tag{D.4.1}$$

In particular test (b) holds on $\mathscr{A}$. This implication uses no two-out-of-three axiom beyond the existence of the relative localization.

Proof. The space of natural transformations between the two constant A-diagrams is

$$\mathrm{Map}_{\mathrm{Fun}(NA,\mathcal{C})}(a_A, b_A) \simeq \mathrm{Fun}(NA, \mathrm{Map}_{\mathcal{C}}(a,b)) \simeq \mathrm{Map}(BA, \mathrm{Map}_{\mathcal{C}}(a,b)).$$

For the first equivalence, express the mapping space as the fibre of the arrow ∞-category over the ordered endpoint pair. The functor $\mathrm{Fun}(NA, -)$ preserves this limit, and the resulting fibre is a space. For the second, a functor to a space factors through groupoid completion, whose underlying space is BA. Taking components and using the fully faithful functor R_A gives (D.4.1). The construction is natural under restriction in A and in both endpoints. Precomposition by a weak equivalence of spaces is an equivalence of mapping spaces, hence induces the required bijection. □

D.5. The second test (b) can fail between finite shapes REFUTED. There are a six-object finite poset X, with all arrows weak, and a weak equivalence $u : Y \to B$ between finite categories, such that $u^* : H_{s,s}(B) \to H_{s,s}(Y)$ is not surjective. Thus the formulation of (b) using only weak equivalence to a Thomason-fibrant category is not equivalent to its constant-diagram formulation, even with finite test categories.

Proof. Let Y have minima r_0, r_1, maxima q_0, q_1, and all four relations $r_i < q_j$. Adjoin to Y two incomparable objects s, n, both above every object of Y, to obtain X. These relations and transitivity give all arrows. The nerves realize respectively a circle and its unreduced suspension, a 2-sphere. Let B be the category with objects a, b and exactly two nonidentity arrows $f, g : a \rightrightarrows b$; there are no composable pairs of nonidentity arrows. Send r_i to a, q_j to b, the edge $r_0 < q_1$ to g, and the other three edges to f. The oriented cycle

$$\gamma = (r_0 q_0) - (r_1 q_0) + (r_1 q_1) - (r_0 q_1)$$

maps to $f - g$. Both realizations are circles and Bu has degree one, so $u \in W_\infty$.

A functor $B \to X$ must send f and g to the same arrow because X is a poset. This identifies the entire functor category, including natural transformations, with $X^{[1]}$. Evaluation at the source is a homotopy equivalence $BX^{[1]} \to BX$: the constant-arrow section has a natural transformation to the identity. Since $BX \simeq S^2$,

$$H_{s,s}(B) = \mathrm{Aut}_{\Pi_1 B(X^{[1]})}(s_s) = 1.$$

Here localization at all arrows is the fundamental groupoid of the nerve; this statement concerns ordinary localization, not an assumed diagram-rectification theorem.

Write $j : Y \hookrightarrow X$ and choose $y_0 = r_0$. In X^Y the natural-transformation loop

$$s_Y \longleftarrow j \longrightarrow n_Y \longleftarrow (y_0)_Y \longrightarrow s_Y \tag{D.5.1}$$

evaluates, on $BY \times S^1$, to the fundamental suspension cycle $n * \gamma - s * \gamma$. The two varying meridians give its two cones; the fixed return meridian contributes no two-simplex. This is a nonzero integral 2-cycle, and NX has no nondegenerate 3-simplices. Its coefficient on each of the eight triangles is $+1$ or -1, with zero boundary. Evaluation and pairing with the fundamental cohomology class of BX therefore define a homomorphism from $\pi_1 B(X^Y)$ at s_Y to $\mathbb{Z}$ which sends (D.5.1) to 1. Its powers are distinct. This proves that $H_{s,s}(Y)$ contains an infinite-order element, proving nonsurjectivity.

The attached program checks the posets and relations, the image of γ, all eight coefficients and their zero boundary, and the absence of three-simplices. It finds 198 objects of X^Y and 18 objects of X^B. This sharpens the companion's ms. 134 calculation by replacing its infinite category $B\mathbb{Z}$ by the finite parallel-arrow category B. □

D.6. What follows formally from the higher loop test PROVED. Each corresponding formula in (c) implies the associated representability in (a). In a pointed ∞-category with the required homotopy pullbacks and pushouts, the choices $\Omega b = 0 \times_b 0$ and $\Sigma a = 0 \amalg_a 0$ satisfy (c), with all loop coherences. Test (b) on finite posets alone does not force the existence of both representing functors.

Proof. Apply π_0 to (D.1.1), using $\pi_0 \Omega K = \pi_1(K, *)$. Conversely, the universal property of the displayed pullback gives $\mathrm{Map}(a, 0 \times_b 0) \simeq * \times_{\mathrm{Map}(a,b)} *$, and the pushout gives the dual formula. These equivalences use the same iterated pullbacks that define loop multiplication, so they respect its coherent structure; the two representabilities yield the adjunction by Yoneda.

For the last assertion use finite pointed simplicial sets with weak homotopy equivalences. This is a finite cofibration category, as in D.11 below, and hence is rectifiable at finite posets by D.7. Thus (b) holds there by D.4. A representing object for ΩS^1 cannot be finite: evaluation at the pointed 0-sphere would give

$$\mathrm{Hom}(S^0, \Omega S^1) = \pi_1 \mathrm{Map}_*(S^0, S^1) = \mathbb{Z},$$

whereas the left side is the finite set π_0 of any finite candidate. The opposite relative category supplies the dual failure for suspension. The equivalence of its localization with the full subcategory of finite CW homotopy types follows directly from simplicial approximation: maps and homotopies between finite polyhedra are represented after finite subdivision, which is a weak equivalence, and subdivision of any finite homotopy is still finite. No conclusion that the disjunctive version of (a) fails is being drawn from this example. □

4.2 Sufficient hypotheses and intrinsic cofibrations

D.7. Rectification theorems with their actual shape bounds KNOWN. The following sufficient conditions concern the canonical comparison R_A, and in fact give equivalences before taking homotopy categories. An ordinary category with weak equivalences and fibrations in the sense of Cisinski rectifies every finite inverse category; its dual rectifies every finite direct category. In particular each rectifies every finite poset. An ABC cofibration category satisfying all six axioms, or its dual, rectifies all categories small relative to the universe of its coproduct and sequential-colimit axioms. The same is true for a Quillen model category complete and cocomplete relative to that universe.

Proof. For finite diagrams the exact localization statement is Cisinski (2019, Theorem 7.6.17):

$$L(\mathrm{Fun}(NI^{\mathrm{op}}, C)) \longrightarrow \mathrm{Fun}(NI^{\mathrm{op}}, LC)$$

is an equivalence for finite direct I and an ∞-category with weak equivalences and fibrations. Apply it to $C = NM$; nerve preserves ordinary diagram categories. Dualizing gives the covariant cofibration assertion. Definition 7.4.12 of that source allows nonfibrant objects and does not assume functorial factorizations. The ordinary finite theory is developed in Cisinski (2010, Definition 1.1, Theorems 1.30 and 2.21, and Corollary 3.20); its coefficient convention is contravariant.

For infinite shapes use Lenz (2018, Theorem 3.33), together with the localization comparison for quasicategories of frames. The word "cofibration category" there includes arbitrary small coproducts and countable composites: Definition 1.5 and Warning 1.7 must be retained. An equivalent canonical formulation is Cisinski (2019, Theorem 7.9.8), with the hereditary hypothesis in Definition 7.9.1; Remark 7.9.7 explains the sufficient sequential-colimit axiom. These are statements about localization of the diagram category, not merely about existence of an abstract equivalent derivator.

For a general model category, its cofibrant objects satisfy the ABC axioms in the stated universe. The cofibrant replacement comparisons for diagram categories used in Lenz's proof identify the induced functor with R_A. The underlying derivator theorem is also Cisinski (2010, Corollaire 6.21), and the model-category case is treated in Cisinski (2003, Théorème 6.11). The result is therefore not restricted to combinatorial or simplicial model categories. It does not assert rectification at diagrams larger than the universe for which the model category has these limits and colimits. □

D.8. Enough intrinsic cofibrations give finite rectification PROVED. Suppose M has an initial object and all pushouts, W contains the isomorphisms and satisfies two out of three, and every map factors as an intrinsic W-cofibration followed by a member of W. Then (M, Cof_W, W) satisfies Rădulescu-Banu's precofibration axioms CF1–CF4 (Rădulescu-Banu, 2009, Definition 1.1.1), equivalently the dual of Cisinski's derivable-category axioms. Consequently (M, W) is rectifiable at every finite direct category, and in particular at every finite poset and every span. Thus, under these hypotheses including the initial object, no counterexample asked for in the POLES addendum can exist at a finite poset.

Proof. An intrinsic cofibration $i : A \to B$ means that the pushout functor $i_! : A\backslash M \to B\backslash M$ preserves underlying W-maps. Isomorphisms have this property. Two successive pushouts identify with pushout along the composite, so these maps compose; the same identification shows stability under arbitrary pushout. The initial object's identity is a cofibration. These are CF1. CF2 is the assumed two-out-of-three property, and CF4 is the given factorization, stronger than its restriction to cofibrant sources.

It remains to prove that a trivial intrinsic cofibration remains weak after arbitrary cobase change. Let $i : A \to B$ lie in both classes and factor a proposed base-change map as $A \xrightarrow{j} A' \xrightarrow{w} A''$ with j intrinsic and $w \in W$. Put $B' = B \amalg_A A'$ and $B'' = B' \amalg_{A'} A''$. Since j is intrinsic and $i \in W$, $A' \to B'$ is in W. Since it is also intrinsic, $B' \to B''$ is in W. In the commuting square with the other vertical arrow $w : A' \to A''$, two out of three now gives $A'' \to B'' \in W$. This proves CF3, with actual comparison maps. It is precisely the needed part of POLES A.9, verified here from its definition rather than imported from a gluing theorem.

D.7 now applies to the nerve of this category with weak equivalences and cofibrations. It includes the replacement of noncofibrant objects, so we have not assumed that every $0 \to X$ is an intrinsic cofibration. In Szumiło's convention (Szumiło, 2014, Section 1.1, axioms C0–C5), one additionally restricts to cofibrant objects and requires the stated two-out-of-six condition; the stronger convention is not needed to invoke Cisinski's theorem. In Lenz's convention one must also verify CF5–CF6. Neither change of convention is implicit in the finite conclusion. □

D.9. Initial objects do not make every object intrinsically cofibrant Refuted. The hypotheses of D.8 do not imply that every object is Cof_W-cofibrant. They also do not imply the infinite coproduct and sequential-colimit axioms CF5–CF6.

Proof. Take the diamond lattice $0 < a, b < 1$ and $W = \{\text{identities}, 0 \to a, 0 \to b\}$. All pushouts are joins and 0 is initial. The two-out-of-three axiom is checked on its only nontrivial composable pairs $0 < a < 1$ and $0 < b < 1$. The complete intrinsic class consists of the identities and $0 \to 1, a \to 1, b \to 1$. Indeed, pushout of $0 \to b \in W$ along $0 \to a$ is $a \to 1 \notin W$, and the dual argument excludes $0 \to b$. Every map into 1 is intrinsic because its pushout sends every object under its source to 1. Every weak map factors as its identity followed by itself; all other nonidentity maps are intrinsic and factor followed by an identity. This verifies every hypothesis of D.8 but excludes cofibrancy of a, b. The complete finite enumeration is attached.

For the infinite axioms take finite sets, with W the isomorphisms. Every map is intrinsic, every factorization can end in an identity, and finite pushouts exist. A countable coproduct of singletons does not exist. Neither does the colimit of the cofibration sequence $\{0\} \subset \{0, 1\} \subset \cdots$. To verify the latter without relying on a set-theoretic picture, a putative finite colimit T would give $\mathrm{Hom}(T, \{0, 1\}) \cong \{0, 1\}^{\mathbb{N}}$, which is impossible. Thus both CF5 and CF6 can fail. This example remains rectifiable, since its weak equivalences are isomorphisms; the failure concerns the proposed axioms, not the conclusion of an unproved converse. □

4.3 Counterexamples separating the proposed criteria

D.10. All cubes do not detect rectification at finite posets Refuted. Rectifiability at [1] and all its finite products does not imply rectifiability at all finite posets. A counterexample has six objects and twelve arrows, including identities.

Proof. Let P be the face poset of the boundary of a triangle, with objects $v_0, v_1, v_2, e_{01}, e_{12}, e_{20}$, relations given by incidence, and $W = \mathrm{Mor}(P)$. Its nerve realizes to a circle, so $L(P, W) \simeq B\mathbb{Z}$, regarded as an ∞-groupoid. For any category A with an initial object a_0, evaluation $\mathrm{Fun}(A, P) \to P$ has a section by constants and a natural transformation from the section

followed by evaluation to the identity. Consequently it induces an equivalence after groupoid completion. Evaluation also gives an equivalence $\mathrm{Fun}(NA, BP) \simeq BP$, since BA is contractible. The square of these two evaluation maps with R_A commutes. Thus R_A is an equivalence. This applies to every cube, including [0].

For $A = P$, the objects on the strict side are monotone maps $P \to P$. The degree of every such map is $-1, 0$, or 1: a simplicial map of the six-edge polygon cannot traverse more than six edges, whereas a map of degree d must traverse at least $6|d|$ edges after cancellations. The target $\mathrm{Ho}\,\mathrm{Fun}(NP, BP)$ has objects of every integral degree. Isomorphic objects there have equal degrees, so the degree-two object is not in the essential image. The exhaustive enumeration in `d_finite_checks.py` finds 234 monotone maps, with degree counts $3, 228, 3$ for $-1, 0, 1$, respectively. This verifies the entire finite search, independently of the length argument. □

D.11. Finite posets versus the infinite cyclic group PROVED. There is a small relative category rectifiable at every finite direct category, hence every finite poset, but not at $B\mathbb{Z}$. There is such an example fibrant in the usual Barwick–Kan model structure. Thus Barwick–Kan fibrancy does not imply rectification at all small shapes.

Proof. Use a small skeleton M of finite simplicial sets, where finite means having finitely many nondegenerate simplices. Give it weak homotopy equivalences and monomorphisms as cofibrations. Finite pushouts remain finite; a pushout of a trivial monomorphism is a weak equivalence. For a map $f : A \to B$, its finite simplicial mapping cylinder factors f as a monomorphism followed by a simplicial homotopy equivalence. All objects are cofibrant. Thus the finite cofibration axioms hold, and D.7 applies. The same statements hold for pointed finite simplicial sets, used in D.6.

Any strict functor $B\mathbb{Z} \to M$ consists of a finite simplicial set K with an actual simplicial automorphism t. Its automorphism group embeds in the permutation group of the finite set of nondegenerate simplices. Therefore t has finite order. In contrast, let X be a finite simplicial model of a wedge of two circles. The Nielsen homotopy equivalence sending the free generators $a \mapsto a$, $b \mapsto ab$ induces, in a chosen basis of H_1,

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad T^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

It gives an infinite-order automorphism of X in $L(M, W)$. To see that this automorphism belongs to the stated localization, triangulate its graph map after a finite subdivision and invert the subdivision map; its homotopy inverse is represented the same way. Equivalently, the usual finite cylinder homotopy calculus gives the homotopy category of finite CW types. A loop representing this equivalence in the core of $L(M, W)$ defines a coherent functor $B\mathbb{Z} \to L(M, W)$.

An equivalence of coherent $B\mathbb{Z}$-diagrams conjugates their monodromies in the homotopy category. After applying H_1, finite-order monodromy remains finite order under conjugacy. It cannot be conjugate to T. This proves failure of essential surjectivity of $R_{B\mathbb{Z}}$. The matrix computation is included in the executable check; its displayed general formula proves infinite order without a finite-test inference.

Finally take M^{op}. This is a fibration category and is Barwick–Kan fibrant by Meier (2016, Theorem 4.13). Localization and diagram comparison commute with opposites, with the indexing category also replaced by its opposite. Since $B\mathbb{Z} \cong (B\mathbb{Z})^{\mathrm{op}}$, the same obstruction applies. For M itself the cited theorem gives fibrancy in the *conjugate* Barwick–Kan structure (Corollary 4.15); it is essential to distinguish the two structures. □

D.12. Rectification does not imply Barwick–Kan fibrancy REFUTED. A nine-object relative poset is rectifiable at every small shape but is not Barwick–Kan fibrant.

Proof. Let F be the fence

$$0 < 1 > 2 < 3 > 4 < 5 > 6 < 7 > 8, \qquad W = \mathrm{Mor}(F).$$

Successively delete the left endpoint of the remaining fence, sending it to its only neighbour. At each step the retraction followed by inclusion is comparable pointwise to the identity, hence admits a natural transformation in one direction. The eight steps give a fixed finite zigzag from id_F to a constant functor. Applying $\mathrm{Fun}(A, -)$ gives such a contraction for every A, with no dependence of the number of steps on A. Thus both $L(F, W)$ and $L(\mathrm{Fun}(A, F), W_A)$ are contractible spaces. The specified comparison R_A is an equivalence.

For a relative category with every arrow weak, the zeroth level of the Barwick–Kan bisimplicial nerve is

$$(N_\xi F)_{0,\bullet} = \mathrm{Ex}^2 NF.$$

This follows directly from double subdivision and $\xi\widehat{[n]} \cong c\,\mathrm{Sd}^2\,\Delta^n$ on underlying categories; see Meier (2016, Example 2.5 and Definition 2.6). An edge of this simplicial set is a monotone map from the five-vertex fence $0 < 1 > 2 < 3 > 4$ to F, with the indicated endpoint values. There are edges $0 \to 4$ and $4 \to 8$, obtained from the consecutive subfences. There is no edge $0 \to 8$: every successive pair of values is equal or adjacent in the undirected nine-vertex fence, so four steps cannot cover distance eight. These two edges define an inner two-horn whose filler would supply the missing edge. Hence $\mathrm{Ex}^2 NF$ is not Kan. A fibrant relative category has a Reedy fibrant N_ξ, so its zeroth level must be Kan. This proves the claim. The program enumerates every five-fence map and verifies both the missing edge and all eight contraction retractions. □

D.13. A relative poset presentation need not rectify REFUTED. Presenting an ordinary category as an ∞-localization of a relative poset does not imply that the naive strict diagram localizations are its coherent diagram categories. The presenting relative poset can be finite.

Proof. Use the six-element P of D.10. Choose an oriented circuit in its nerve. Define $q : P \to B\mathbb{Z}$ to send one incidence edge to the generator (or its inverse according to orientation) and all other incidence edges to $0 \in \mathbb{Z}$. There are no composable nonidentity pairs, so these assignments define a functor. Its map on classifying spaces has degree one. Since every arrow maps to an isomorphism and BP is a circle, q exhibits $B\mathbb{Z}$ as the ∞-localization of $(P, \mathrm{Mor}(P))$.

Every strict $B\mathbb{Z}$-diagram in a poset is constant: its generator must act by an automorphism, and posets have only identity automorphisms. Therefore $P^{B\mathbb{Z}} \cong P$. Its localization is a connected groupoid. On the coherent side, $\mathrm{Fun}(B\mathbb{Z}, B\mathbb{Z})$ has components indexed by the homomorphisms $\mathbb{Z} \to \mathbb{Z}$, that is, by integers. The comparison lands in the component of the trivial homomorphism. It is not essentially surjective. It already fails at the finite poset P by D.10.

There is also a presentation of a *finite* category for which rectification fails at that same finite category. Let K be the infinite join of copies of S^0, indexed by $\mathbb{N}$. Its nonempty faces are finite sets $F \subset \mathbb{N} \times \{+, -\}$ containing at most one sign at each index. Simultaneous sign reversal acts freely. Let Q be the poset of its face orbits, with $[F] \leq [G]$ if $F \subseteq G$ after a choice of signs. Nonemptiness makes the aligning sign unique. A chain of face orbits therefore lifts uniquely after one representative is chosen, so

$$NQ \cong N(\mathrm{Face}\,K)/C_2, \qquad BQ \simeq \mathbb{RP}^\infty \simeq BC_2.$$

Here K is contractible: its finite joins are spheres of dimensions tending to infinity, so every homotopy group of the union vanishes; the CW Whitehead theorem applies. The action is free, and the quotient is the displayed classifying space.

Normalize each face representative so that its least-index sign is $+$. For $[F] \leq [G]$ let $q([F] \leq [G]) \in C_2$ be the unique sign ε with $\varepsilon F \subset G$. Signs multiply under composition, so $q : Q \to BC_2$ is a functor. It classifies the free cover of face representatives and induces the displayed equivalence after groupoid completion. All arrows are marked. But $Q^{BC_2} \cong Q$, whereas $\mathrm{Fun}(BC_2, BC_2)$ has two components, indexed by the trivial and identity homomorphisms of C_2. The canonical comparison misses the identity component. The target category and the test shape each have one object and two arrows; their complete composition law and homomorphisms are included in the finite check.

The general existence of a relative poset presentation is Lurie (2026, Theorem 6.3.6.1, tag 02MD). Its universal property concerns *coherent* functors out of the presenting poset that invert the marked arrows. It does not identify localization of the ordinary category of strict functors into that poset. The two variances and the placement of the localization are different. □

4.4 Kaledin's intrinsic axioms and the localization they represent

D.14. The precise shapes and axioms in Kaledin v3 KNOWN. In Kaledin (2026), Pos means small posets of finite chain dimension; it does not mean finite posets. The notation pos denotes finite posets. The category Pos^+ consists of left-bounded posets, and $\mathrm{Pos}^{\pm}$ of left-finite posets. Left-bounded means that every principal lower set has finite chain dimension; left-finite means that every such set is finite.

A restricted enhanced category is a small additive, semiexact, separated, nondegenerate reflexive family over Pos, satisfying excision and the cylinder axiom. An enhanced category is the analogous family over Pos^+ with semicontinuity added. These are Definitions 7.2.1.1, 7.2.1.3 and 7.2.1.4, using Definitions 7.1.1.1–7.1.1.2, 7.1.1.9 and 7.1.4.6; the shapes are Definitions 2.1.5.3–2.1.5.4. The representing complete Segal space in Theorem 7.3.1.3 can be chosen κ-small when the family is κ-bounded for an uncountable regular κ.

Source audit and interpretation. The axioms have distinct content. Nondegeneracy says that evaluation at all points is conservative. Reflexivity specifies the adjunctions associated with $J \to J \times [1] \to J$. Separatedness asks that the induced functor $C_{J\times[1]} \to \mathrm{Fun}([1], C_J)$ be full and essentially surjective. Additivity identifies the value on a disjoint union with the product of the values, including infinite unions allowed by the shape universe. Semiexactness requires an epivalence for the prescribed standard pushout square. Excision, cylinder and semicontinuity require semicartesianness for precisely the squares (6.3.2.1), (6.3.2.3), (6.3.2.2), respectively. They are conditions on the family and its transition functors, not conditions on its value at the point alone.

Proposition 7.2.3.7 extends a *good reflexive family* on $\mathrm{Pos}^{\pm}$ to a bar-invariant ample subcategory containing $\mathrm{Pos}^{\pm}$; it preserves the additional listed axioms. It does not extend an arbitrary family on finite posets. In Lemma 7.2.4.1 the naive localization of a model category on left-finite posets has an enhanced extension; its identification with the naive family on Pos^+ is asserted when the model category is complete. This extra clause must not be discarded. Chapters 6 and 7 of v3, dated 23 February 2026, remain marked preliminary by the author; we treat the stated representation theorem as a preprint theorem, not as a separately re-proved or refereed result. □

D.15. Intrinsic recognition on Kaledin's shape category KNOWN. Fix compatible

universes and a small relative category (M, W). Let $\mathcal{D}_J = M^{J^{\mathrm{op}}}[W_J^{-1}]$ with its actual restriction functors. The following are equivalent:

1. This family on Pos satisfies the restricted enhanced-category axioms in D.14.
2. The canonical comparisons

$$\mathcal{D}_J \longrightarrow \mathrm{Ho}\,\mathrm{Fun}(NJ^{\mathrm{op}}, L(M, W))$$

   are equivalences for every $J \in \mathrm{Pos}$.

The analogous equivalence holds on Pos^+ with the full enhanced-category axioms. In particular, the category represented by a naive enhanced family is the specified localization $L(M, W)$, not an unidentified ∞-category. This is a formal consequence of Kaledin's theorem and the localization universal property; it is not claimed as a new independent recognition theorem.

Proof. Write $\mathcal{K}(X)_J = \mathrm{Ho}\,\mathrm{Fun}(NJ^{\mathrm{op}}, X)$, and use the associated Grothendieck fibration. The identification of this notation with Kaledin's $K(X)$ follows from the expansion in Lemma 7.3.1.1, formulas (7.3.1.7)–(7.3.1.8): its core at J is the fundamental groupoid of the derived space of NJ^{op}-diagrams, and expansion recovers their homotopy category. For ordinary M, this is exactly the represented family $K(M)$ of (7.1.1.5) and Example 7.2.1.16.

We need the following elementary two-categorical localization fact. The pointwise localization map

$$\ell : K(M) \longrightarrow \mathcal{D}$$

has fully faithful precomposition on categories of cartesian functors into any category-valued family. In each fibre this is the ordinary localization universal property, including natural transformations: a transformation between two functors on the localized category is uniquely determined on the original category, and naturality for inverses follows from naturality for the inverted arrows. The compatibility isomorphisms for restrictions descend in the same way. Their coherence equations hold because they can be checked before localization. This proves the asserted fact for cartesian functors and modifications.

Assume (1). Theorem 7.3.1.3 identifies $\mathcal{D}$ with $K(X)$ for a complete Segal space X. By Definition 7.2.2.1 an enhanced functor is exactly a cartesian functor over the shape category; it need not preserve Kan extensions. The same theorem, applied to ℓ, gives $f : NM \to X$ with $K(f) \cong \ell$. At the point, every $w \in W$ is sent to an isomorphism in $\mathrm{Ho}\,X$, so f inverts W. Put $C = L(M, W)$, with $q : NM \to C$. Its universal property gives $g : C \to X$ and an equivalence $gq \simeq f$.

The canonical diagram comparison $\theta : \mathcal{D} \to K(C)$ commutes with restriction, hence is an enhanced functor. The theorem represents it by $h : X \to C$, with $hf \simeq q$. Consequently $hgq \simeq q$. Precomposition with q is fully faithful on ∞-functor categories, so $hg \simeq \mathrm{id}_C$. On the other side,

$$K(g)\theta\ell \cong K(gq) \cong K(f) \cong \ell.$$

The pointwise localization fact proved above gives $K(g)\theta \cong \mathrm{id}_{\mathcal{D}}$. The last clause of Theorem 7.3.1.3, which identifies homotopies with isomorphisms of enhanced functors, now gives $gh \simeq \mathrm{id}_X$. Thus h is an equivalence, and its represented functor is the *specified* comparison θ. This proves (2).

Conversely the canonical comparisons form a cartesian equivalence of families. The coherent family $K(C)$ is enhanced by Lemma 7.3.1.1, and the axioms are invariant under such equivalences. The argument for Pos^+ is identical, using the corresponding clauses of Theorem 7.3.1.3. No two-out-of-three assumption is used in this formal argument beyond whatever relative-category convention is chosen for forming the localization. □

D.16. An explicit infinite-poset failure of additivity PROVED. For the six-object relative poset $(P, \mathrm{Mor}(P))$ of D.10, the naive family fails Kaledin's additivity axiom at the countable discrete poset $I = \mathbb{N}$. This shape has chain dimension zero and is left-finite. In contrast, the finite category E of D.2 satisfies every enhanced-category axiom: there is no failed axiom to assign to that example.

Proof. Since every arrow is weak, the fibre at I is $\Pi_1 B(P^I)$. Choose a vertex $a \in P$ and consider the comparison on its constant object:

$$\mathrm{Aut}_{\Pi_1 B(P^I)}(a^I) \longrightarrow \prod_{n\in\mathbb{N}} \mathrm{Aut}_{\Pi_1 BP}(a) = \mathbb{Z}^{\mathbb{N}}.$$

Every morphism on the left is represented by a finite zigzag of arrows in the product category. If it has length r, every coordinate loop traverses at most r edges of the six-edge circle. Its coordinate degrees are therefore uniformly bounded in absolute value, for example by r. The element $(n)_{n\in\mathbb{N}}$ of the product cannot be in the image. Thus the additivity functor $\mathcal{D}_I \to \prod_I \mathcal{D}_{\mathrm{pt}}$ is not full. This is exactly Definition 7.1.4.6, not merely a failure of an unspecified enhancement property. It also exhibits failure of rectifiability at this particular infinite poset.

For E and identity weak equivalences, the family is $K(E)$. Example 7.2.1.16 states and verifies every axiom for $K(E)$ for *any* ordinary category E, by the fact that the prescribed squares are pushouts in Cat. Finite products or finite limits in E are not required. The failure of a product $x \times x$ in D.3 therefore concerns completeness, not enhancement.

The extension theorem cannot repair this example while retaining its naive values on all left-finite posets, since I already belongs to that class. More generally, when a good left-finite family has an enhanced extension, uniqueness of that extension does not assert that an independently defined naive family on larger posets is the extension. D.15 characterizes when the latter identification holds. □

4.5 The later manuscript classes and the remaining boundaries

D.17. What the later cofibration and Quillen axioms imply PROVED. The following implications hold, with the displayed size assumptions.

1. The axioms Cof_0–Cof_3 of XVII ms. 39–40 and XIX ms. 33–36, with the stated initial object and the pushouts required by the axioms, give rectification at every finite direct category. They do not require every object to be cofibrant.

2. A quillénisable pair in the sense of XIII ms. 50–51 is rectifiable at finite direct categories. Already M0, M2, M′ and M″ at XIII ms. 7–10 supply enough intrinsic cofibrations and enough intrinsic fibrations.

3. Adding the full transfinite axiom Cof_4 of XIX ms. 33, including existence of the indicated small filtered colimits, gives the ABC coproduct and sequential-colimit axioms. Rectification then holds at all shapes small in that universe, and the left Kan extensions asked for in XIX Question 3.14, ms. 48, exist. No small generating set or functorial factorization is needed for this conclusion.

4. More specifically, enough intrinsic cofibrations, an initial object, pushouts, small filtered colimits, and preservation of W by such filtered colimits imply (3).

Proof. For (1), Cof_0 is two out of three with isomorphisms; Cof_1 is closure of cofibrations under composition and cobase change; Cof_2 is closure of trivial cofibrations under cobase change; Cof_3 factors every map as a cofibration followed by a weak equivalence. Together with the initial object these are CF1–CF4, so D.7 applies. The source at XVII ms. 40 explicitly restricts the Brown category to the full subcategory of cofibrant objects and compares its localization with that of the original category. The same replacement, included in Cisinski's theorem, avoids an unjustified all-objects-cofibrant assumption.

For (2), M0 supplies finite limits and colimits. One M2 factorization has first map in $C \subset \mathrm{Cof}_W$ by M″ and second in $TF \subset W$ by M′. The other has first map in $TC \subset W$ and second in $F \subset \mathrm{Fib}_W$. Apply D.8 and its dual. The definition of "exact weak Quillen structure" further asks for $W = TF \circ TC$ (M2 bis); it must not be substituted without proof for the stronger identities $TC = C \cap W$, $TF = F \cap W$. For finite rectification this extra exactness is unnecessary. Likewise the word "quillénisable" by itself does not supply colimits of unbounded diagrams. XIII ms. 53 explicitly discusses examples without small filtered colimits.

For (3), the initial object and finite pushouts give finite coproducts. A small coproduct is the filtered colimit of its finite subcoproducts, so it exists. To verify CF5 for a family of cofibrations, well order the family and replace one summand at each successor step. Each step is a cobase change of that cofibration; at a limit take the prescribed filtered colimit. Transfinite closure of C proves that the resulting coproduct map is a cofibration. If every map was trivial, use transfinite closure of $C \cap W$. Coproducts of cofibrant objects are cofibrant by the same argument starting at the initial object. Cof_4 applied to a countable sequence supplies CF6, including its assertion for trivial cofibrations. Hence D.7 applies in the ABC convention, including the associated left derivator and its left Kan extensions.

For (4), a transfinite composite of intrinsic cofibrations is intrinsic. Given a weak map under its source, push it out at each stage. At successor stages it remains weak by the defining property of the relevant intrinsic cofibration; at limit stages it remains weak by the assumed filtered-colimit stability. Pushout commutes with the colimit, so the last comparison is precisely the required pushout map. If each stage is also weak, the transfinite composite is weak by the same induction, applying filtered-colimit stability to the diagram from the constant initial stage. D.8 identifies trivial intrinsic cofibrations and supplies the remaining finite axioms. This proves Cof_4 and then (3).

These conclusions also locate two overstatements to avoid. The Brown assertion at XIX ms. 35 about ambient intrinsic cofibrations must be read with the restriction explicitly present at XVII ms. 40, or with the additional properness hypothesis $C \subset \mathrm{Cof}_W$. Further, the functorial factorization in XIX Corollary 3.3 alone gives objectwise factorizations of diagram categories; it does not add CF5–CF6. Our all-shape conclusion uses their verified transfinite source, not functoriality as a replacement for infinite colimits. □

D.18. Accessible localizers on categories KNOWN. For every accessible fundamental localizer W, the pair $(\mathrm{Cat}_{\mathcal{U}}, W)$ is rectifiable at all $\mathcal{U}$-small categories and has the associated strong derivator. It has enough intrinsic cofibrations. Enough intrinsic fibrations, in contrast, is equivalent to right properness of its Thomason model structure, or to generation of W by maps $A \to e$. The pointed versions of (a) and (c) hold in its pointed model category; (b) holds before pointing as well.

Proof. Cisinski (2006, Théorème 5.2.15) constructs the cofibrantly generated left proper Thomason model structure for every accessible W. Apply D.7 and the model-category derivator theorem cited there. Its model cofibrations preserve weak equivalences under every cobase change, so they are intrinsic and its first factorization gives enough intrinsic cofibrations. The same source's Théorème 6.1.11 identifies properness with generation by maps to a point;

Théorème 5.2.15 identifies this with model right properness. Right proper model fibrations are intrinsic, giving the stated factorization.

For clarity the converse has a short Brown argument, which also checks the POLES E.2 step used here. If intrinsic fibrations are enough, their trivial members are stable under all pullbacks by the dual proof of D.8. Every object of Cat is intrinsically fibrant, since a fundamental localizer is stable under products. Let W^{asph} be generated by the W-aspherical categories. For a trivial intrinsic fibration $T \to B$, its pullbacks over B/b are weak equivalences with W-aspherical targets, so T/b is W-aspherical. Theorem A for W^{asph} implies $T \to B \in W^{\mathrm{asph}}$. Brown's lemma, applied to the identity functor from this category of fibrant objects, gives $W \subset W^{\mathrm{asph}}$; the reverse inclusion is definitional. The cited properness theorem now applies.

No initial-terminal identification is implicit: ordinary Cat is not pointed. The model category of pointed objects supplies that identification, and its finite homotopy limits and colimits give D.6. Thus the pointed tests are correctly typed. This result does not infer accessibility merely from a coefficient descent axiom, nor does it impose right properness to obtain rectification. □

D.19. Descent does not replace two out of three Refuted. For the represented derivator of sets, the class W^1_{Set} defined by fully faithful restriction on locally constant objects fails two out of three. Therefore D.7–D.9 and D.17 cannot be applied to $(\mathrm{Cat}, W^1_{\mathrm{Set}})$ merely from its descent properties. Ordinary localization, D.2, D.4 and the formal argument of D.15 do not require this extra hypothesis.

Proof. Let $i : e \to BC_2$ and $r : BC_2 \to e$, so $ri = \mathrm{id}$. Locally constant set-valued diagrams on BC_2 are C_2-sets. The functor r^* gives the trivial action and is fully faithful. The functor i^* forgets the action and is not full: on the regular two-element C_2-set it sees four endofunctions, of which only two are equivariant. Thus $r, ri \in W^1_{\mathrm{Set}}$ but $i \notin W^1_{\mathrm{Set}}$. The multiplication and all four functions are checked in the accompanying finite program. This is a failure of a specific axiom, not an obstruction to forming the relative localization in the first place. □

D.20. Prederivator recovery and the scope of the comparison Known. The references named in Kaledin's introduction are Kevin Arlin, *A higher Whitehead theorem and the embedding of quasicategories in prederivators*, and Daniel Fuentes-Keuthan, Magdalena Kędziorek and Martina Rovelli, *A model structure on prederivators for* $(\infty, 1)$*-categories*. Their embedding and image results are not criteria on the value $M[W^{-1}]$ alone. They concern the family of shapes with its transformations. In particular, homotopy categories of coherent diagrams cannot be replaced by ordinary diagram categories in the homotopy category.

Verified scope and a minimal obstruction. Arlin (2020, Theorems 19, 29 and 37) distinguishes strict and pseudonatural morphisms. Bicategorical full faithfulness for small quasicategories is stated for all small category shapes; the finite-shape assertion is conservativity for a specified map, not the same fullness assertion. Fuentes-Keuthan et al. (2019, Theorems 2.16, 3.1 and 3.4 in the arXiv version) characterize prederivators arising on the nose from quasicategories and construct a model structure on prederivators with strict natural transformations. This distinction from equivalence of naive families is retained in D.15.

For the concrete obstruction take the arrows $S^0 \to *$ and $* \to S^1$. A map between these objects in the coherent arrow category consists of the endpoint maps together with a homotopy of the resulting square. Its mapping space is the homotopy pullback of

$$* \longrightarrow \mathrm{Map}(S^0, S^1) \longleftarrow S^1,$$

where the right map is the diagonal and the left map is a fixed constant pair. Its components form $\mathbb{Z}$: the long exact sequence gives the cokernel of $\mathbb{Z} \xrightarrow{\Delta} \mathbb{Z}^2$. Thus there are infinitely many maps in $\operatorname{Ho}\operatorname{Fun}([1], \mathcal{S})$. In $\operatorname{Fun}([1], \operatorname{Ho}\mathcal{S})$ the corresponding set has one element. The indexing arrow [1] is already sufficient to detect the lost homotopy of a square. □

D.21. An upgrade when all Kaledin product tests are available PROVED. Fix a category A. If $R_{A\times J}$ is an equivalence for every $J \in \mathrm{Pos}$, with the stated compatible restriction functors, then the canonical ∞-functor

$$L(M^A, W_A) \longrightarrow \operatorname{Fun}(NA, L(M, W))$$

is an equivalence. The same holds using Pos^+. This is a sufficient upgrade for a specified comparison; it does not replace Pos by finite posets without proof.

Proof. The hypothesis identifies the naive family of (M^A, W_A) with the enhanced family represented by $X = \operatorname{Fun}(NA, L(M, W))$, since iterated coherent functor categories satisfy the exponential identity. Apply the proof of D.15 to this relative category. That proof shows that the representing object X, with its specified map from $N(M^A)$, is its ∞-localization. This is exactly the displayed functor. Only the map with this specified provenance is covered. □

D.22. Exact remaining rectification and test questions OPEN. The following targets are not settled in this package.

1. An intrinsic necessary-and-sufficient criterion using only finite-poset shapes, or only finite-category shapes, or the shapes of cardinality less than a prescribed regular κ, has not been deduced. D.15 is a criterion on Kaledin's Pos or Pos^+ and cannot be silently restricted to any of these smaller classes.

2. The version of the prompt's mapping-space upgrade which assumes only the equivalences $R_{A\times J}$ for finite posets J remains open here. D.21 establishes the larger shape version. The missing step is to identify the naive finite-shape family of M^A with an enhanced family, or to prove the canonical ∞-functor is fully faithful by a separate finite-test theorem that applies to this source.

3. Whether Barwick–Kan fibrancy alone forces rectifiability at every finite poset, or at every finite category, is not established here. D.11 refutes the implication for all small shapes, already at the countable shape $B\mathbb{Z}$; D.12 refutes the converse even for all small shapes.

4. Enough intrinsic cofibrations *without* an initial object are not covered by D.8. The given least-vertex Reedy arguments in POLES do not themselves supply the missing latching coproducts for arbitrary finite diagrams. The initial-object version of the requested counterexample is impossible by D.8; the unrestricted version without that object is left open here.

5. The exact equivalence in IX-080 between a suitably bounded Cat hot representability formulation of (b) and its restriction-invariance formulation is not proved merely by a Thomason fibrant replacement. D.5 refutes the interpretation that only asks for a weakly equivalent fibrant category. The implication from the manuscript's path-object version of (c) to (b), in the absence of the comparison hypotheses in Parts B–C, remains open. D.6 proves (c)⇒(a) and the actual mapping-space identities in a pointed ∞-category; D.4 proves rectification⇒(b).

6. Whether every nonaccessible fundamental localizer gives the unrestricted derivator on Cat is not settled. D.18 uses accessibility at the model-structure existence step, so its proof gives no answer after removing that hypothesis.

What is and is not supplied by the preceding results. For an uncountable regular κ, an ABC structure with coproducts of fewer than κ summands and the countable composition axiom is a size-controlled sufficient condition for shapes of size less than κ. The constructions in D.7 use finite strings of arrows and countable realization stages; their index sets remain smaller than κ. This explains the necessary universe qualification for a model category. Merely having finite limits and colimits corresponds to a different assertion: a finite category with endomorphisms can have infinitely many nondegenerate simplices in its nerve. It is not covered by the finite-direct theorem.

D.10 rules out the proposed cubical criterion, D.11 separates finite posets from $B\mathbb{Z}$, and D.13 rules out the naive use of relative poset presentations. None is a classification of all finite-shape rectifiable relative categories. Likewise the finite example at IX ms. 23 fails finite completeness, not (a), (b), (c), rectification, or enhancement. It cannot establish any of these still-missing converses. We state these as unresolved targets of this package, without claiming that the entire literature has proved them independent or left them open. □

D.23. The cofibration–fibration asymmetry for every homology theory PROVED. For every nonzero spectrum E, the fundamental localizer $W_E = \{u \mid E \wedge \Sigma^\infty_+ Bu \text{ is an equivalence}\}$ does not have enough intrinsic fibrations. This checks the generality of POLES C.4–C.5, including nonconnective spectra; the sufficient cofibration and rectification statements in D.18 do not require this stronger fibration property.

Proof. First suppose $A \to *$ is an E-equivalence. The space A is nonempty and connected. Indeed, point inclusions and projection to components split off one copy of E for each component; the fold map on two copies has nonzero kernel in some $\pi_n E$. If $\pi_1(A, a)$ had a nonzero homomorphism to $\mathbb{Z}$, divide by its image to obtain a surjection. A loop representing 1 and the classifying map to S^1 would give pointed maps $S^1 \to A \to S^1$ whose composite is homotopic to the identity. Thus ΣE would be a retract of the zero reduced spectrum $E \wedge \Sigma^\infty A$, a contradiction.

Let $F = \langle x, y\rangle$ and $X = BF \simeq S^1 \vee S^1$. Every subgroup of a free group is free, as follows from its covering graph, and every nontrivial free group surjects onto $\mathbb{Z}$. Therefore every homomorphism $\pi_1(A, a) \to F$ is trivial. Since BF is a 1-type, $\mathrm{Map}_*(A, BF)$ is contractible, and evaluation gives $\mathrm{Map}(A, BF) \simeq BF$, with the equivalence given by constant maps.

For any space X the class

$$V_X = \{u \mid \mathrm{Map}(B\,\mathrm{cod}(u), X) \to \mathrm{Map}(B\,\mathrm{dom}(u), X) \text{ is an equivalence}\}$$

is a fundamental localizer. Two out of three and retracts are immediate, and categories with terminal objects are contractible. For relative Theorem A, use the natural decompositions of BA and BB as the homotopy colimits of their comma categories over the common base C. Mapping to X turns these homotopy colimits into homotopy limits, so equivalences on every comma term give the asserted equivalence. The preceding paragraph shows that V_{BF} contains every generator $A \to e$ of W_E^{asph}.

Define $\alpha : F \to F$ by

$$\alpha(x) = x, \qquad \alpha(y) = yxyx^{-1}y^{-1}.$$

Its exponent-sum matrix is I_2, so on the reduced suspension spectrum of the two-circle graph it is the identity matrix. Consequently $B\alpha \in W_E$ for every spectrum E. It is not surjective:

the surjection $\rho : F \to S_3$, $x \mapsto (12)$, $y \mapsto (13)$, sends both $\alpha(x)$ and $\alpha(y)$ to (12). Thus $\rho\alpha(F)$ has two elements, whereas $\rho(F)$ has six. The program verifies the complete S_3 multiplication table, the word, and these two subgroup sizes.

If precomposition by $B\alpha$ were an equivalence of $\mathrm{Map}(BF, BF)$, surjectivity on components would give r with $rB\alpha \simeq \mathrm{id}$. Injectivity, applied to $(B\alpha r)B\alpha \simeq B\alpha$, would then give $B\alpha r \simeq \mathrm{id}$. This would make $B\alpha$ a homotopy equivalence, contrary to nonsurjectivity on π_1. Hence $B\alpha \notin V_{BF}$ and $W_E \neq W_E^{\mathrm{asph}}$. The accessibility-free Brown direction in the proof of D.18 shows that enough intrinsic fibrations would force equality. This contradiction completes the proof without using a right-properness test along an unverified fibration. □

4.6 The small-index obstruction and the audit of the cardinal Kan–Thurston argument

D.K1. Acyclic maps are epimorphisms in the unpointed homotopy category PROVED. Let $t : X \to Y$ be a map of small CW spaces with nonempty, connected, integrally acyclic homotopy fibres. For every small space Z, $t^* : [Y, Z] \to [X, Z]$ is injective. More precisely the codiagonal $c : Y \cup^h_X Y \to Y$ is a weak equivalence. This verifies GALOIS B.9 in the exact variance needed for CATHOT E.17; the codiagonal characterization also appears in Buchholtz et al. (2025, Theorem 2.9).

Proof. For connected Y, the fibre sequence implies that X is connected and $\pi_1 t$ is onto. Realize the homotopy pushout by the double mapping cylinder P, with its two inclusions $j_0, j_1 : Y \to P$. Van Kampen identifies $\pi_1 P$ with $\pi_1 Y *_{\pi_1 X} \pi_1 Y$, and the codiagonal induces an isomorphism to $\pi_1 Y$ because both maps out of $\pi_1 X$ are the same surjection. An acyclic map is a homology isomorphism with every local system on its target (Hausmann and Husemoller, 1979, Definition/Proposition 1.2). For a local system L on Y, compare the Mayer–Vietoris sequences for $Y \leftarrow X \to Y$ and $Y \leftarrow Y \to Y$. The two outer maps are identities and the middle map is t, so $H_*(P; c^*L) \to H_*(Y; L)$ is an isomorphism. Taking $L = \mathbb{Z}[\pi_1 Y]$ and using the established fundamental-group isomorphism identifies this with the homology comparison of the universal covers. Homological Whitehead makes c a weak equivalence.

Since $cj_0 = cj_1 = \mathrm{id}_Y$, it follows that $[j_0] = [j_1]$. A homotopy $at \simeq bt$, for $a, b : Y \to Z$, gives an actual map $h : P \to Z$ with $hj_0 = a$ and $hj_1 = b$. Hence $[a] = [b]$. For disconnected spaces the construction and argument apply to each component; acyclicity gives a bijection on components of X and Y. Maps from a disjoint union of spaces are taken componentwise, with the homotopies assembled on that disjoint union. This includes the empty case and makes no interchange of π_0 with infinite products of ordinary categories. □

D.K2. The envelope construction used by GALOIS B.10 PROVED. For every small group H and every infinite small cardinal $\lambda \geq |H|$ there is an embedding $H \hookrightarrow A$, where A is nonabelian, simple, integrally acyclic, and has cardinality exactly λ.

Proof. We check the two constructions in the supplied proof. First enlarge H to $H * F_\lambda$. For every pair $g \neq 1, h$ of elements of a current group L of cardinality λ, add a different free generator $t = t_{g,h}$ and form

$$w = gtg^{-1}t^{-1}gt^2g^{-1}t^{-2}.$$

The words w and wh have infinite order in $L * \langle t \rangle$. The first is cyclically reduced with eight alternating syllables; cyclic reduction of the second has eight syllables unless $hg = 1$, in which case it has six. In all cases its length is at least two and every displayed free-product syllable is nonidentity. Therefore the two infinite cyclic subgroups can be identified by an HNN extension with relation $s^{-1}ws = wh$. The simultaneous HNN embedding theorem preserves the old group

(Higman et al., 1949, Theorems I–II). The relation puts h in the normal closure of g. Countably repeating this step gives a simple overgroup of cardinality λ: any nonidentity g and any h eventually occur together and satisfy that normal generation condition. Noncommutativity follows from the embedded free group.

For the acyclic envelope, start with $L \times L$ and its two inclusions $j(h) = (h, 1)$, $k(h) = (1, h)$. Add stable letters d, s with

$$d^{-1} j(h) d = j(h)k(h), \qquad s^{-1} j(h) s = k(h).$$

The maps identify embedded copies of L, so HNN again gives an embedding of $L \times L$. Call the resulting group $m(L)$. The homology calculation is the mitosis calculation of Baumslag et al. (1980, Proposition 4.1 and Theorem 4.2). Explicitly, let $f : G \to L$ induce zero in positive degrees below n over a field. In degree n the Künneth decomposition of the diagonal following f has only its two axis terms, since all mixed positive-degree terms vanish. In $m(L)$ each axis map and the diagonal map are conjugate to j. Hence on $H_n(G)$, $(jf)_* = (jf)_* + (jf)_*$, so $(jf)_* = 0$. Induction shows that a map from one stage to its n-th successor in the iteration of m kills positive homology through degree n. Each bar cycle uses finitely many group elements. The countable union is therefore acyclic over every field. Rational coefficients make its positive integral homology torsion; the universal coefficient sequence in the next degree with $\mathbb{F}_p$ coefficients excludes p-torsion for every prime. Its positive integral homology is zero. The construction does not increase the infinite cardinal λ.

Alternate simple and acyclic envelopes, always by embeddings, and take their countable union A. Given $g \neq 1$ and $h \in A$, both lie in a later simple stage, so the normal closure of g in A contains h. Each finite integral bar cycle lies in a later acyclic stage and bounds there. Thus A has all the stated properties. This verifies the cardinal assertion as well as the homological assertion, rather than obtaining a merely countable special case from a citation. □

D.K3. Verification of the cardinal-controlled Kan–Thurston construction PROVED. For every connected $\mathcal{U}$-small CW complex Y and every $\mu \in \mathcal{U}$ there are a $\mathcal{U}$-small group G and an acyclic map $t : BG \to Y$ such that $\pi_1 t$ is onto and every homomorphism $G \to L$ with $|L| \leq \mu$ kills $P = \ker(\pi_1 t)$. In particular the proposed (KT_μ) is true with the quantifiers in the prompt. The induction in GALOIS B.11, including its subcomplex-injectivity step, proves it.

Proof. We give the checked induction in full, since injectivity is the material issue in the inherited proof. Choose a small simplicial complex K and a homotopy equivalence $h : |K| \to Y$ (Hatcher, 2002, Theorem 2C.5). Choose an infinite $\lambda \in \mathcal{U}$ exceeding μ and at least the number of simplices of K. The construction makes, simultaneously for every subcomplex $L \subseteq K$, a space $T(L)$ and an actual map $t_L : T(L) \to |L|$. The choices attached to each simplex are made once and reused in every subcomplex containing it. The induction assertions are that $T(L)$ is aspherical on each component and that every inclusion $T(L') \hookrightarrow T(L)$ is injective on fundamental groups, component by component.

In dimensions zero and one put $T(L) = |L|$ and $t_L = \mathrm{id}$. Reduced edge paths prove both assertions for graphs. Suppose the construction is made through dimension $n-1$, where $n \geq 2$. For every n-simplex σ choose by D.K2 an embedding

$$\pi_1 T(\partial\sigma) \hookrightarrow C_\sigma$$

into a simple acyclic group of cardinality λ. The connected aspherical source admits a cellular map $a_\sigma : T(\partial\sigma) \to BC_\sigma$ realizing this embedding. Let Q_σ be its mapping cylinder, with $T(\partial\sigma)$ at the free end. Thus this end is a CW subcomplex, its map to Q_σ is injective on π_1, and $Q_\sigma \simeq BC_\sigma$ is aspherical and acyclic. Define

$$T(L) = T(L^{(n-1)}) \cup_{\coprod_{\sigma \in L_n} T(\partial\sigma)} \coprod_{\sigma \in L_n} Q_\sigma. \tag{D.K3.1}$$

On a cylinder, t_L moves $t_{\partial\sigma}(x)$ linearly to the barycentre of σ, and it sends BC_σ to that barycentre. This defines the required continuous map, compatible with all subcomplex inclusions.

We use the following graph-of-spaces fact with its precise injectivity hypothesis. A graph of connected aspherical CW vertex spaces, with aspherical edge spaces attached by mapping cylinders and injective edge homomorphisms, is aspherical and embeds its vertex groups. The amalgamated-product and HNN normal forms give the embeddings. In its universal cover, the incidence graph is the tree of cosets. Every vertex and edge inverse image is contractible. The space over any finite subtree is contractible by successively attaching a contractible vertex along a contractible edge cylinder. A sphere map and any finite cellular data lie over a finite subtree, so the universal cover is weakly contractible and hence contractible. This is also the amalgamation argument recalled in Baumslag et al. (1980, Section 1, before Proposition 1.1). Components of a disconnected edge space are treated as separate edges.

In (D.K3.1) each boundary group injects into the lower-skeleton group by induction and into C_σ by construction. Thus $T(L)$ is aspherical, and the lower skeleton and each BC_σ embed on π_1. For the stronger subcomplex assertion, fix $L \subseteq K'$ of dimension at most n and form

$$V = T((K')^{(n-1)}) \cup_{T(L^{(n-1)})} T(L). \tag{D.K3.2}$$

This is a graph of spaces with one edge for each component of $T(L^{(n-1)})$. Its two edge inclusions are injective: the first by the dimension-$n-1$ induction hypothesis, the second by the preceding application to $T(L)$. Therefore $T(L) \to V$ and $T((K')^{(n-1)}) \to V$ are injective on fundamental groups. Attach the remaining Q_σ for $\sigma \in K'_n \setminus L_n$. Their boundary groups inject through $T((K')^{(n-1)})$ and keep doing so after each attachment. The result is exactly $T(K')$, by the same pushout construction and the choices fixed simplexwise. Hence $\pi_1 T(L) \to \pi_1 T(K')$ is injective. This proves the required step without treating an arbitrary map of aspherical spaces as injective. Filtered unions over dimensions preserve these assertions because spheres, nullhomotopies, and group relations use finitely many cells.

For every local system M of abelian groups on $|L|$ we claim

$$H_*(T(L); t_L^* M) \xrightarrow{\cong} H_*(|L|; M). \tag{D.K3.3}$$

It holds for graphs. In an n-simplex the coefficient system is constant. The map of pairs $(Q_\sigma, T(\partial\sigma)) \to (\sigma, \partial\sigma)$ is an isomorphism on relative homology by the long exact sequences, acyclicity of Q_σ, and induction for the boundary. Excision gives the direct sum of these relative comparisons over all n-simplices of L. The skeletal long exact sequence proves (D.K3.3) at that dimension; cellular chains with local coefficients commute with the union over skeleta. Thus it holds without a dimension bound. All spaces constructed have at most λ cells: bar models of the groups do, and the finite products and countable unions involved preserve that bound.

Put $G = \pi_1 T(K)$ and let N be the normal closure in G of all the embedded C_σ, $\dim\sigma \geq 2$. The map t_K kills these groups. Its induced quotient map is the isomorphism

$$G/N \xrightarrow{\cong} \pi_1 |K|. \tag{D.K3.4}$$

Indeed, at dimension two, killing C_σ kills exactly the boundary relator of that simplex. For $n \geq 3$, the inductive quotient assertion for $\partial\sigma$ and $\pi_1 \partial\sigma = 0$ say that the boundary group is normally generated by the already attached lower cone groups. It introduces no new relation after those groups have been killed. Van Kampen proves (D.K3.4) by induction and by the union over dimensions. Each C_σ is perfect, so their normal closure N is perfect: its generating conjugate subgroups lie in $[N, N]$.

Choose a based homotopy equivalence $b : BG \to T(K)$ and set $t = ht_K b$. Equation (D.K3.3), with Hausmann and Husemoller (1979, Definition/Proposition 1.2), makes this specified map acyclic. Equation (D.K3.4) identifies its perfect kernel with N. A homomorphism $G \to L$, $|L| \le \mu$, restricts trivially to each C_σ: simplicity makes a nontrivial restriction injective, which its cardinality $\lambda > \mu$ forbids. It therefore kills their normal closure. When K has dimension at most one there are no cone groups and the kernel is trivial, so that case is included. □

D.K4. The small-index version of CATHOT E.17 PROVED. Let $\mathcal{H} = \mathrm{Cat}\,\mathrm{hot}_{\mathcal{U}}$, whose morphisms are functors modulo finite zigzags of natural transformations. Let I be a nonempty $\mathcal{U}$-small filtered category and $K : I \to \mathcal{H}$. If

$$H_K(X) = \mathrm{colim}_{i \in I} \mathrm{Hom}_{\mathcal{H}}(X, K_i)$$

inverts W_∞ on all $\mathcal{U}$-small categories X, then it inverts W_1. Moreover the specified unit

$$(K_i)_{i \in I} \longrightarrow (\Pi_1 K_i)_{i \in I}$$

is an isomorphism in $\mathrm{Ind}_{\mathcal{U}}(\mathcal{H})$, without changing the indexing category. This verifies GALOIS B.12 and settles the numbered target CATHOT E.17 in that small-index scope.

Proof. The functor $q : \mathcal{H} \to \mathrm{Hot}$ is localization at nerve weak equivalences. A natural transformation supplies a functor on $X \times [1]$; the two endpoint inclusions become equal after its projection to X is inverted. Thus passing from Cat to $\mathcal{H}$ before localization has no effect. The assumed H_K is therefore the pullback of a presheaf H' on Hot, and its natural comparison is

$$H'(Y) \cong \mathrm{colim}_{i \in I} [Y, BK_i]. \tag{D.K4.1}$$

To justify this crucial step without assuming each K_i local, take presheaves in a larger universe. Restriction q^* is fully faithful. Its left adjoint, left Kan extension along q^{op}, sends a representable to the corresponding representable on Hot by Yoneda, and preserves colimits. Its unit is invertible on the presheaf already in the image of q^*. Applying it to the displayed colimit proves (D.K4.1), including its naturality and its specified comparison map.

Choose a small infinite cardinal μ bounding the numbers of arrows of every K_i. Such a uniform bound exists because I and the family are small in one universe. For every component Y_c of a small CW space Y apply D.K3 to obtain $t_c : BG_c \to Y_c$. Put $T = \coprod_c BG_c$, $t = \coprod_c t_c$, and let $v : Y \to \tau_{\le 1} Y = \coprod_c B\pi_1 Y_c$ be first truncation in the homotopy category. The quotient homomorphisms $G_c \twoheadrightarrow \pi_1 Y_c$ represent vt. Write $\mathbf{B}G$ for the one-object groupoid of a group G, so that BG denotes its classifying space. For every i restriction induces an actual isomorphism of ordinary categories

$$\mathrm{Fun}\left(\coprod_c \mathbf{B}\pi_1 Y_c, K_i\right) \xrightarrow{\cong} \mathrm{Fun}\left(\coprod_c \mathbf{B}G_c, K_i\right). \tag{D.K4.2}$$

A functor on $\mathbf{B}G_c$ is an object k together with a homomorphism $G_c \to \mathrm{Aut}_{K_i}(k)$. Its target group has size at most μ, so it factors uniquely through $\pi_1 Y_c$. A natural transformation is an arrow commuting with these actions; because the quotient homomorphism is surjective, this is exactly the same condition after factorization. Thus (D.K4.2) is an isomorphism on objects and arrows, even when K_i has noninvertible endomorphisms and when there are infinitely many components.

By (D.K4.1), the resulting bijection is the composite

$$H'(\tau_{\le 1} Y) \xrightarrow{H'(v)} H'(Y) \xrightarrow{H'(t)} H'(T).$$

The second arrow is injective by D.K1: at each stage $[Y, BK_i] \to [T, BK_i]$ is injective, and a filtered colimit of these compatible injections remains injective. The composite is bijective by (D.K4.2), so the first arrow is bijective. Here taking components was done after an isomorphism of functor categories; no invalid commutation with an infinite product is involved. For $Y = BX$ this says exactly that $H_K(\Pi_1 X) \to H_K(X)$, induced by the category groupoid-completion unit, is a bijection. Since Π_1 takes W_1 to equivalences of groupoids, H_K inverts W_1.

Finally, groupoid completion reflects $\mathcal{H}$ into the full subcategory of groupoids modulo natural isomorphisms. The ind-Hom formula

$$\mathrm{Hom}_{\mathrm{Ind}(\mathcal{H})}((A_i), (B_j)) = \lim_i \mathrm{colim}_j \mathrm{Hom}_{\mathcal{H}}(A_i, B_j)$$

extends that reflection to ind-objects. Write $LK = (\Pi_1 K_i)$ and $\eta : K \to LK$. Taking the limit over i of the bijections $H_K(\Pi_1 K_i) \to H_K(K_i)$ supplies $r : LK \to K$ with $r\eta = \mathrm{id}_K$. The object LK has the same orthogonality to every $X \to \Pi_1 X$, since each groupoid target does. Precomposition with η is therefore injective on endomorphisms of LK. The identity $(\eta r)\eta = \eta = \mathrm{id}_{LK}\eta$ gives $\eta r = \mathrm{id}_{LK}$, proving that the specified unit is an ind-isomorphism. The empty test category is included: both sides of (D.K4.2) are then terminal. □

D.K5. Application to V ms.44–46 and the universe restriction PROVED. For a small relative category (M, Σ) and fixed objects a, b, the countable path system of V ms. 44–45 is in the scope of D.K4. Consequently, if

$$X \longmapsto \mathrm{Hom}_{M^X[\Sigma_X^{-1}]}(a_X, b_X)$$

inverts weak equivalences of all small X in that same universe, it factors through Π_1. This conclusion does not require that the functor first be represented by a space. If it is represented by a space Z, then Z is a 1-type. No such conclusion follows from invariance on a class of shapes bounded relative to a larger M.

Proof. Every localization morphism in M^X is represented by a finite word of arrows and formal inverses of pointwise Σ-arrows. A single word has one finite length and type for the whole diagram X, however large X is. Padding by identities puts it at a stage K_n of the countable alternating path system. Commutative ladders with identity endpoint components give the same localization morphism: after inverting the backward arrows, the commutative-square equations telescope from the first endpoint to the second. No vertical arrow is inverted in this calculation, so arbitrary vertical components are allowed. Conversely, the relations defining localization, namely composition, identities, and inverting a Σ-arrow, are represented by finite reparametrizations and such ladders. A finite proof of equality uses a common later stage. Applying this description in the functor category gives the natural bijection of sets

$$\mathrm{Hom}_{M^X[\Sigma_X^{-1}]}(a_X, b_X) \cong \mathrm{colim}_{n\in\mathbb{N}}\, \pi_0\, \mathrm{Fun}(X, K_n).$$

This is exactly the variance and countable indexing printed at V ms. 45. It can also be obtained by applying C.P3 to the relative category (M^X, Σ_X), taking connected components, and using the literal exponential identification of its fixed-type path category with $\mathrm{Fun}(X, K_n)$. Each K_n is small because M is small, so D.K4 applies. Neither two-out-of-three nor saturation is needed for this formal localization description.

If $H(X) = [BX, Z]$, its first-truncation invariance implies that $[\tau_{\le 1} Y, Z] \to [Y, Z]$ is bijective for every small space Y, by categorical models of spaces. At $Y = Z$ the identity factors through $Z \to \tau_{\le 1} Z$. Thus Z is a retract in the homotopy category of a 1-type, and its homotopy groups above degree one vanish. In a rectified diagram theory the constant-to-constant formula identifies Z with the relevant mapping space, giving the stated restriction on the unbounded small presentation.

The proof chooses groups of size strictly larger than a uniform bound on the stages. Those classifying categories are still small in the original universe, but they need not belong to a prescribed bounded class of admissible shapes. A model category large relative to those shapes does not supply the uniform bound used in D.K4. This is the precise point at which the bounded formulation avoids the obstruction. The expression "Thomason–Kan" at V ms. 46 can accordingly be read as localness in $\mathrm{Cat}\,\mathrm{hot}$, realized here by groupoid stages; Thomason fibrancy alone does not justify that reading. □

5 What the categorical path construction supplies

5.1 The all-arrows theorem and the distinction between its models

E.1. The verified all-arrows comparison KNOWN. Let X be a small category and let J be the integer fence with arrows $2i \to 2i \pm 1$. Write $\mathcal{P}(X)$ for the category of functors $J \to X$ constant with identity arrows outside a finite interval, and $p = (\sigma, \beta) : \mathcal{P}(X) \to X \times X$ for the eventual endpoints. Then the constant-path section $X \to \mathcal{P}(X)$ is a weak equivalence, the one-endpoint fibres are weakly contractible, and the exact two-endpoint fibres are homotopy fibres. The canonical comparison span between these fibres and the all-type fibres $\mathrm{Ch}(X; a, b)$ consists of weak equivalences and respects composition in Hot. Thus

$$B\,\mathrm{Ch}(X; a, b) \simeq B\mathcal{P}(X; a, b) \simeq P_{a,b}(BX).$$

The actual even translation on $\mathcal{P}(X; a, b)$ is the identity in Hot. The actual one-sided constant-tail maps q_L, q_R are weak equivalences, including for any wide relative class on the corresponding relative fibres.

Verification and precise use of the supplied proofs. We checked CATHOT E.20–E.25 and the companion's blocks at VII ms. 166–169, including the variance of the lifting categories. In accordance with the task, we cite those checked proofs rather than reproduce the all-arrows argument. The essential hypotheses are positive-length supports when invoking a finite-stage Grothendieck fibration, both endpoint identities on fibre morphisms, and filtered unions on objects and arrows. The all-type endpoint functor has only the smoothness used in the repaired argument; its properness is false, as E.3 records.

The shorter q_L argument in the supplied second review is valid. Let $\mathcal{P}_{\leq 2k}$ be the full subcategory of paths constant on $[2k, \infty)$. Its union is $\mathcal{P}$, and after translation each successive inclusion is the collar-shift map S of CATHOT E.23. The explicit stationary-on-finite-support collar homotopy shows $BS \simeq 1$. A sequential union of these weak equivalences is a weak equivalence because the nerve and homotopy groups after Ex^∞ commute with filtered colimits. This applies on the exact fibres and preserves the relative backward-arrow condition. It does not give a bounded zigzag of natural transformations; the counterexample in Part C shows why that stronger conclusion is false. □

E.2. A strict categorical multiplication on the loop model PROVED. For every $x \in X$, the all-type category $\Omega_{\mathrm{cat}}(X, x) = \mathrm{Ch}(X; x, x)$ is a strict monoidal category, natural in pointed functors $(X, x) \to (Y, y)$. Its nerve is a simplicial monoid, its component monoid is the group $\pi_1(BX, x)$, and its classifying space has the weak homotopy type of $\Omega_x BX$. Its multiplication agrees with the usual loop multiplication in Hot. The construction uses no pushout-product axiom and no fibrant replacement of X.

Proof. Use concatenation and the empty path, as in C.1. A pointed functor applies to every vertex, arrow and natural-transformation component; it preserves concatenation and the empty path as equalities. The component identification follows from the path-word presentation of the groupoid completion. The space and its multiplication are identified by E.1, whose cited companion block treats composable pairs over X^3, so it identifies this operation and not just an abstract homotopy type. All constructions here use finite products, concatenation, interval collapses and filtered unions. No categorical pushout along a product of endpoint inclusions occurs.

Strict associativity supplies an A_∞ multiplication on this model. The strict composition-compatible weak equivalence ρ of B.2 identifies it as an A_∞ loop space: here all arrows are weak, so the weak-vertical and full categories coincide, and the localization is the space BX. This higher identification uses B.2, while E.1 by itself supplies compatibility only in Hot. □

For clarity, we fix the lifting-category convention. For $f : E \to S$, $e \in E$, and $\alpha : f(e) \to s$, an outgoing lift is an arrow $h : e \to e'$ with $f(e') = s$ and $f(h) = \alpha$; an arrow from h to h' is $k : e' \to e''$ with $f(k) = 1_s$ and $kh = h'$. For $\alpha : s \to f(e)$, an incoming lift is $h : e' \to e$ over α, with arrows $k : e' \to e''$ over 1_s satisfying $h'k = h$. In the W_∞ sense, properness means that every outgoing lift category is weakly contractible, and smoothness means the analogous incoming condition. These are the lifting-category criteria of Maltsiniotis (2005, Definition 1.1 and Theorem 1.2), with duality fixing the variance. A Cartesian lift would in particular be terminal in its incoming lift category; a coCartesian lift would in particular be initial in its outgoing lift category.

E.3. The all-type endpoint functor is not a strict bifibration Refuted. For $X = [1] = (0 \xrightarrow{w} 1)$, the functor $p : \mathrm{Ch}(X) \to X \times X$ is smooth in the W_∞ lifting-category sense, but it is not proper and is neither a Grothendieck fibration nor a Grothendieck opfibration. In particular the duality argument asserting properness of this total category is false.

Proof. Smoothness is the incoming-lift contraction checked in the companion, VII ms. 164: an incoming lift is compared with a slice over the path obtained by adjoining the two given endpoint arrows; that slice has a terminal object. This works with arbitrary type maps in their stated direction.

For failure of properness take the empty path at 0 and the outgoing base arrow $(1_0, w) : (0, 0) \to (0, 1)$. An outgoing path morphism lies above a type map $e \to \tau$. Endpoint preservation forces $\tau = e$. Such a target cannot have endpoints $(0, 1)$. The outgoing lift category is empty, which also excludes an opfibration.

For failure of a Grothendieck fibration take the empty path at 1 and the incoming base arrow $(w, 1_1) : (0, 1) \to (1, 1)$. Its lift category is exactly $\mathrm{Ch}([1]; 0, 1)$: a type collapse to e is unique and the transformation to the constant path 1 is unique. A Cartesian lift would be a terminal object c in this category. Write τ for its type. Since its endpoints differ, τ is nonempty. Let d be the path of type $\tau\tau$ which is constant at 0 on the first block and is c on the second. There are two different type projections $\tau\tau \to \tau$, collapsing respectively the first and the second block. For the first projection the pullback of c is exactly d. For the second it is c on the first block and constant at 1 on the second. Pointwise inequalities give a natural transformation from d to this second pullback; at the endpoints they are identities. Thus $d \to c$ has two distinct morphisms, contradicting terminality.

The program checks all 543 such [1]-paths of length at most six and constructs these two morphisms; the argument proves the claim at every length. The finite base category has exactly two objects, its two identities and w, so all its composition data have been specified. Taking the opposite of a Grothendieck construction reverses its indexing category. Reversing a path inside a fixed Ty does not do this. That is why the claimed deduction of properness cannot be repaired by merely reversing X. □

E.4. Both homotopical lifting properties do not give Cartesian lifts REFUTED. The eventual alternating endpoint functor $\mathcal{P}(X) \to X \times X$ is smooth and proper for every X. Even for $X = [1]$ it is neither a Grothendieck fibration nor a Grothendieck opfibration.

Proof. The two homotopical lifting properties are precisely the even-even and odd-odd finite-support arguments checked in E.1. The relevant lift categories are filtered unions of categories with a terminal, respectively initial, object. Filtered unions preserve weak contractibility; they need not preserve those universal objects.

For [1], consider incoming lifts into the constant path 1 over $(w, 1)$. Such a lift is an eventual path from 0 to 1, with its unique transformation to 1. A terminal lift d cannot exist: choose a single-transition path which changes from 0 to 1 across a direct edge so far to the left that d is still 0 just after that edge. Pointwise order then excludes a transformation from this path to d. Dually, outgoing lifts from the constant path 0 over $(1, w)$ are the same eventual paths. For any proposed initial lift, choose a single-transition path which changes to 1 later than that lift's eventual-1 tail begins. The required arrow to the later-transition path is impossible. The proof uses the two-object finite category just checked in E.3 and paths at arbitrarily large supports; a finite cut-off alone would miss the obstruction. □

E.5. A functorial categorical mapping-path factorization PROVED. For every functor $f : A \to X$, define

$$E(f) = A \times_{X,\sigma} \mathcal{P}(X), \qquad i_f(a) = (a, \mathrm{const}_{f(a)}), \qquad p_f(a, c) = \beta(c).$$

Then $f = p_f i_f$, the functor $i_f : A \to E(f)$ is in W_∞, and $p_f : E(f) \to X$ is smooth and proper. Its exact fibres compute the homotopy fibres of Bf. This factorization is natural in strictly commuting squares of functors. It is not asserted to be a Thomason cofibration–fibration factorization.

Proof. Smoothness and properness are stable under base change and composition: their incoming or outgoing lift categories for a pullback are the original lift categories, and the composite criterion follows by taking the appropriate comma construction. Thus the map $E(f) \to A \times X$, a base change of p, is smooth and proper, as is its composite with the projection $A \times X \to X$.

We justify the weak-equivalence step before using the section. A functor with both lifting properties has fibre-to-comma weak equivalences and satisfies Quillen's Theorem B, as in the checked proof CATHOT E.21. The same proof applies after every base change. It follows that any base-change square is homotopy Cartesian: compare its strict fibres with the derived fibres, and then apply the fibrewise weak-equivalence criterion for spaces over the same base. This includes empty fibres and empty bases. For $\sigma : \mathcal{P}(X) \to X$ all strict fibres are weakly contractible by E.1. Every base change of σ is therefore a weak equivalence. In particular the projection $r : E(f) \to A$ is one. Since $r i_f = 1_A$, two-out-of-three makes i_f a weak equivalence. The homotopy-fibre description for p_f, and the equation $p_f i_f = f$, give the assertion about Bf.

For a commuting square $(u, v) : f \to f'$, apply u to a and apply v pointwise to the path; this defines the functor $E(f) \to E(f')$ and commutes strictly with both displayed structure maps. The usual categorical cocylinder factorization has the same purpose; Shoikhet (2016, Theorem 2.8) is an exact reference for Evrard's construction with its own indexing convention. We have not identified those conventions by an unproved isomorphism. The present proof needs no right properness of another chosen model structure and no product-of-cofibrations axiom. □

E.6. The standard eventual path object does not solve the five-element bifibration test REFUTED. Let $R = \{1, e_{00}, e_{01}, e_{10}, e_{11}\}$ with

$$e_{ij}e_{kl} = e_{il}, \qquad 1r = r1 = r,$$

and let $X = \mathbf{B}R$. Then $\mathcal{P}(X) \to X \times X$ is neither a Grothendieck fibration nor a Grothendieck opfibration. The all-type functor is not an opfibration either. These failures do not refute existence of some other strict bifibration representing the path local system over X.

Proof. Set $e = e_{00}$. Every product er and re is nonunit. Let c be the constant diagram, whose edges are all 1. An incoming lift of $(1, e)$ into c consists of a path d and a transformation $\alpha : d \to c$ with eventual components 1 on the left and e on the right. For an even vertex i and an adjacent odd vertex j, its naturality equation is

$$\alpha_i = \alpha_j d_{i\to j}.$$

For any sufficiently large odd m, there is such a lift with components 1 for $i \le m$ and e for $i \ge m + 1$; its sole nonidentity edge is e on the arrow $m + 1 \to m$. Given any proposed terminal lift α, choose a coordinate where $\alpha_i = e$, and choose m beyond it. A morphism from this delayed lift γ to α would require $\alpha_i\theta_i = \gamma_i = 1$, which is impossible. Thus no terminal incoming lift exists.

For outgoing lifts $\alpha : c \to d$ the equation is $d_{i\to j}\alpha_i = \alpha_j$. A delayed step at an even m, with edge e on $m \to m + 1$, supplies another lift. If α were initial, the arrow to a sufficiently delayed lift would require $\theta_i\alpha_i = 1$ at a coordinate where $\alpha_i = e$, again impossible. Finally an outgoing morphism from the empty path in the all-type category must have empty target; its two endpoint components coincide. Hence the base arrow $(1, e)$ has no such lift.

The program checks all 125 associativity equations in R, all its nonunit-ideal equations, and the two families of naturality equations. Fiedorowicz's unnumbered lemma on pp. 33–34 identifies $BX \simeq S^2$ (Fiedorowicz, 2002); we opened and checked its five-element presentation. Our failure of strict lifts uses only the displayed multiplication table. It therefore addresses the precise standard path models in the STRATA E.11 test case, while leaving that existence question open. □

5.2 The limit of the interval argument

E.7. The pushout-product property fails already for the shortest alternating block REFUTED. For every $n \ge 1$, let $\Phi_n = (0 \to 1 \leftarrow 2 \to \cdots \leftarrow 2n)$. The pushout-product

$$\alpha_n : (\{0\} \times \Phi_n) \amalg_{\{0\}\times\{0,2n\}} ([1] \times \{0, 2n\}) \longrightarrow [1] \times \Phi_n$$

is a weak equivalence of finite categories but is not an intrinsic W_∞-cofibration. For $n = 1$ there is an explicit weak equivalence in its undercategory whose cobase change has source contractible and target of homotopy type S^3.

Proof. Put $P_n = [1] \times \Phi_n$ and write $l_i = (0, i)$, $u_i = (1, i)$. The source A_n of α_n is the full sieve containing all l_i and only u_0, u_{2n}. Its nerve is a tree, whereas BP_n is contractible. Thus $\alpha_n \in W_\infty$.

We specify the pushout rather than treating it as a homotopy pushout. Let C have objects x, z, a, b, arrows $\lambda, \mu : x \rightrightarrows z$, $r : z \to a$, $v : z \to b$, and the equal composites

$$r\lambda = r\mu = \eta : x \to a, \qquad v\lambda = v\mu = \theta : x \to b.$$

There are no other nonidentity arrows. Its full subcategories on x, z, a and on x, z, b are cones on the parallel-arrow circle. Their nerves cover NC and intersect in that circle, so $BC \cong S^2$. Let $D = C^{\triangleright}$, with new terminal object s. Define $F : A_n \to C$ by

$$F(l_0) = z, \quad F(u_0) = b, \quad F(w) = a \text{ at every other object.}$$

The arrows $l_0 \to l_1$ and $l_0 \to u_0$ go to r, v, and all other nonidentity arrows of A_n go to identities. The composite $A_n \to D$ is a weak equivalence since both categories are contractible.

For a sieve pushout $Q = P_n \amalg_{A_n} D$, the objects are those of D and the remaining u_k, $1 \le k \le 2n-1$. Its crossing arrows are the explicit coends

$$\mathrm{Hom}_Q(d, u_k) = \int^{w \in A_n} \mathrm{Hom}_D(d, Fw) \times \mathrm{Hom}_{P_n}(w, u_k).$$

There are no arrows from an upper complementary object to D. For $k = 1$ the predecessor diagram maps finally to the fork $z \to a, z \to b$. For $n = 1$ it has one extra object u_2 attached by $l_2 \to u_2$, which does not change that finality. The fork inclusion into C is ordinary final: at x its comma category is the connected four-edge square, and at the other objects it has an initial object. The crossing set is consequently a singleton for every $d \in C$, and empty for $d = s$. For $k \ge 2$ the predecessor diagram is nonempty connected and constant at a, so the crossing presheaf is $D(-, a)$. The upper transition to u_1 gives the unique map from that representable presheaf to the stated singleton presheaf on C.

It follows that Q retracts onto the full subcategory $C \cup \{s, u_1\}$ by sending u_k, $k \ge 2$, to a. The arrows represented by 1_a give a natural transformation from the inclusion after this retraction to 1_Q. The retract consists of two terminal cones on C, with no arrow between their tips. Therefore

$$BQ \simeq \Sigma BC \simeq S^3.$$

Cobase change along α_n has taken the weak equivalence $A_n \to D$ to $P_n \to Q$, which is not a weak equivalence. This is exactly failure of the defining preservation condition for an intrinsic cofibration.

The program constructs the coend equivalence relations, all arrows and all compositions, and checks every associativity equation for $n = 1, 2, 3$. For $n = 1$, P_1 has $(6, 15)$ objects/arrows and Q has $(6, 20)$; the normalized rational Betti numbers of Q are $(1, 0, 0, 1, 0)$. The cone proof establishes its integral homotopy type; numerical ranks alone would not do so. The supplied POLES H.6 states the obstruction for $n \ge 2$. The extra predecessor at $n = 1$ has been handled above, so the shortest block is included here. This strengthens the checked range without using the missing POLES computation files. □

E.8. Finite interval cotensors have the right total type but need not have the right strict fibres PROVED. For every finite oriented interval I, the constant-diagram map $X \to \mathrm{Fun}(I, X)$ is a categorical homotopy equivalence, natural in X. This does not make its endpoint evaluation a homotopy path fibration. In particular, the strict fibre of $\mathrm{Fun}([1], [1]) \to [1] \times [1]$ at $(1, 0)$ is empty, whereas the topological path space between those vertices is contractible and nonempty.

Proof. Order the interval's vertices geometrically as $0, \dots, r$. The functors $v_k(i) = \min(i, k)$ join the constant-at-0 functor to 1_I by natural transformations oriented according to the successive edges. Precomposition gives the desired finite zigzag on $\mathrm{Fun}(I, X)$, relative to evaluation at 0. Naturality in X is immediate. For $I = X = [1]$ the indicated strict fibre consists of arrows $1 \to 0$, of which there are none. On realization $B[1]$ is an interval, and its endpoint path space is nonempty contractible. Thus the total equivalence cannot be restricted

to arbitrary strict fibres. The eventual alternating construction in E.1 supplies precisely the additional relative result. Cotensoring and concatenation require no pushout-product axiom; invoking a monoidal model axiom for them would add a hypothesis not used in their proofs. □

5.3 Other localizers and the proposed stratified interpretation

E.9. Weak-equivalence preservation under pullback depends on the localizer REFUTED. The W_∞-smooth and proper path functors cannot in general be called universally weak-equivalence-preserving fibrations for an arbitrary larger fundamental localizer. This fails for rational homology equivalences, even over the one-object category $\mathbf{B}C_2$.

Proof. Put $X = \mathbf{B}C_2$, with its unique object x, and let $q : \mathcal{P}(X; x, ?) \to X$ be endpoint evaluation. Its total category is weakly contractible by E.1, and its strict fibre is of type ΩBC_2, which has two contractible components. The functor $e \to X$ induces an isomorphism on rational homology. Indeed positive-degree group homology of a finite group with rational coefficients vanishes by the averaging contraction, and BC_2 is connected. In the pullback square along q, the induced map from the strict fibre to the contractible total category is not a rational homology equivalence: it maps $\mathbb{Q}^2$ to $\mathbb{Q}$ on H_0. The base category and group law are completely specified by $C_2 = \{1, g\}$ and $g^2 = 1$.

This is a failure of universal preservation of a chosen larger weak-equivalence class. It does not contradict the contractibility of the incoming and outgoing lift categories, or the space-valued homotopy-fibre theorem. Those are the senses of smoothness, properness and homotopy fibration used in E.1–E.5. Any coefficient-specific claim needs its own definition and hypothesis. □

E.10. Components of Σ do not generally define the proposed exit-path stratification REFUTED. For an arbitrary relative category (M, Σ), there need not be a stratification of BM whose strata are the components of the groupoid completion of Σ and whose exit-path mapping spaces are either of the proposed relative path models.

Proof. Take the one-object category with endomorphism monoid $\{1, e\}$, $e^2 = e$, and $\Sigma = \{1\}$. This is a saturated two-out-of-three class: its localization is the same ordinary category, and e is not invertible. There is only one proposed stratum. The classifying space is contractible, since the natural transformation with component e joins the identity functor to the constant functor sending every arrow to 1. A one-stratum exit-path category is its fundamental ∞-groupoid, whose mapping space here is contractible. The relative localization instead has the discrete two-element endomorphism space $\{1, e\}$. Part B's isomorphism-only case gives the same two components for both path models. Hence neither can be that one-stratum exit-path space.

The obstruction is structural: within a stratum all morphisms of an exit-path category are invertible, while a relative localization can retain noninvertible endomorphisms. Positive statements for posets or suitable EI categories need additional hypotheses and an actual stratified comparison. The finite directed test with two parallel arrows has two components when only isomorphisms are inverted; inverting both arrows gives the ordinary circle path space with components indexed by $\mathbb{Z}$. This is consistent with the endpoint cases, but it does not justify the general stratification proposal. □

E.11. Existence of a different strict bifibration in the rectangular-band case OPEN. For $X = \mathbf{B}R$ of E.6, we do not settle whether the path local system is represented by some strict bifibration over X with weakly contractible total category and fibres of type ΩBX. E.6 rules out the standard eventual endpoint construction as such a bifibration, and rules out the

all-type endpoint construction as an opfibration. It leaves the existence question STRATA E.11 intact.

The remaining step. For bases consisting entirely of monomorphisms or entirely of epimorphisms, the supplied STRATA construction uses an injectively fibrant diagram and adjoints to transition functors on categories of bounded contractible objects. In the rectangular-band base the four nonunits have neither cancellation property. Our path constructions give contractible lifting categories, not universal lifts. Replacing those categories by equivalent ones cannot be assumed to produce both adjoints coherently over the monoid. The missing step is precisely such a functorial, two-sided adjoint transition system or an obstruction to its existence. □

E.12. Contractible origin-path categories need not be filtered or totally aspherical KNOWN. For $Y = (z \to a, z \to b)$ and origin z, the exact all-type category $P(Y, z)$ is contractible in Cat hot, but is neither filtered nor totally W_∞-aspherical. More generally $P(Y, y_0) \to Y$ is ordinary final when BY is connected and simply connected.

Verification of TOTASPH G.1–G.6. We checked the supplied proof with its exact type maps. For an initial origin, replacing every path by the constant path of the same type gives natural transformations to the identity and to the empty-path constant functor; hence the asserted two-step contraction. The paths ending at a and at b have no common successor, because their endpoint images would need one in Y.

For finality, an object of $y\backslash P(Y, y_0)$ is moved into the exact fibre over y by appending its structure arrow backwards and collapsing that last edge. If BY is simply connected this fibre is nonempty connected by the groupoid-completion computation of components. For this particular Y, its full inclusion in the four-object category C of E.7 is ordinary final, with the connected square as comma category at x. Composing the two final functors gives a final image of $P(Y, z)$ whose classifying space is S^2. This violates total asphericity. The finite category C and its complete composition table are independently checked in the program for E.7. Thus XIIba-C005 is false, with no claim that contractibility alone is insufficiently verified. □

E.13. Finite set-valued tests are not faithful REFUTED. Choose a small skeleton $\mathcal{A}_{\mathrm{fin}}$ of the full subcategory of Hot spanned by finite categories (finitely many objects and arrows). The restricted set-valued Yoneda functor

$$R : \mathrm{Hot} \longrightarrow \mathrm{Fun}(\mathcal{A}_{\mathrm{fin}}^{\mathrm{op}}, \mathrm{Set}), \qquad R(Y)(A) = \mathrm{Hom}_{\mathrm{Hot}}(A, Y) \cong [BA, BY],$$

is not faithful. An actual endofunctor of a small one-object category induces a map distinct from the identity in Hot but the identity on every one of these test sets. Thus the faithfulness argument in VIIc-C004 is false, although the particular equality $T^2 = 1$ is established independently in E.1.

Proof. We independently verify the finite-support argument of CATHOT F.2 (Supplied AI research corpus, 2026). Let G be the group of finite-support permutations of $\mathbb{N} = \{0, 1, \ldots\}$ and define $s : G \to G$ by

$$s(\sigma)(0) = 0, \qquad s(\sigma)(i + 1) = \sigma(i) + 1.$$

This homomorphism induces the actual endofunctor $\mathbf{B}s : \mathbf{B}G \to \mathbf{B}G$ of the one-object category $\mathbf{B}G$; write $Bs : BG \to BG$ for its realization. Its image fixes 0, whereas the transposition (0 1) does not. Thus s is not surjective, so Bs is not a homotopy equivalence and cannot equal 1_{BG} in the homotopy category.

Every component of the nerve of a finite category has finitely generated fundamental group, since its one-skeleton is finite. If P is a finitely generated group and $\rho : P \to G$ is a homomorphism, the supports of the images of a finite generating set have finite union. Choose m with this union contained in $\{0, \dots, m\}$. Conjugation by the finite cycle $c = (0\ 1\ \cdots\ m+1)$ satisfies

$$c\rho(p)c^{-1} = s(\rho(p)) \qquad (p \in P).$$

For a connected CW complex Z, maps to the 1-type BG are classified by homomorphisms $\pi_1 Z \to G$ modulo conjugation. Indeed a based map is determined by its homomorphism on the one-skeleton, extends over the two-cells precisely when the relations hold, and extends uniquely up to homotopy over higher skeleta because the higher homotopy groups of BG vanish; forgetting the basepoint introduces conjugation. Consequently $Bs \circ f \simeq f$ for every $f : BA \to BG$ with A connected and finite. Apply this separately on every component for arbitrary finite A; the empty test has one map. Hence $R(\mathbf{B}s) = R(1_{\mathbf{B}G})$, although the maps are distinct in Hot. The counterexample category $\mathbf{B}G$ is infinite; the argument treats every finite-category test. □

A Disposition of manuscript targets

This index records the disposition of the requested manuscript rows and numbered targets. "Proved under hypotheses" refers only to the setting specified in the cited statement; an accompanying "Open" retains the broader literal path assertion. In particular, the comparison problem B.8 is not resolved by a theorem about the restricted hammock category or by a rectification theorem for diagram categories.

Target	Disposition	Exact scope and numbered location
Chapter IX: composition, pointed structure, and linear structure		
IX-037	Proved	Strict all-type composition and its coherent enrichment are C.1; the specified sequential operation is associative and unital in Hot by C.P4. Its localization interpretation is proved for weak verticals by B.2, and for the all-arrows loop model by E.2. General J remains B.8.
IX-040	Known under comparison; Open otherwise	C.5 identifies Quillen's groupoid with the fundamental groupoid of a derived function complex, for cofibrant source and fibrant target or specified replacements. Identification with the literal full path model requires the endpoint-compatible comparison retained in B.8. The conclusion is an equivalence of groupoids, with specified vertices.
IX-041	Known under comparison; Open otherwise	Derived mapping groupoids include all zigzag components, as explained in C.5; B.2 supplies the restricted hammock realization. The corresponding identification of every literal arbitrary-vertical path vertex, with its naturality, retains the comparison requirement. A.P1 establishes the component statement unconditionally.
IX-044	Proved under hypotheses; Open generally	C.3 gives the smash-product correction and unit S^0. C.4 constructs strictly absorbing pointed composition on cofibrant-fibrant objects of a pointed simplicial model category. The specified stabilized-path lift, with its absorption coherences, remains C.10.

Continued on the following page.

Target	Disposition	Exact scope and numbered location
IX-049	Proved under hypotheses; Open generally	Associative unital smash composition is proved in the pointed model of C.4. The exact categorical path pairings, including the proposed stronger assertion in pointed Cat hot, remain C.10; an unpointed equivalence supplies neither the absorption homotopies nor their compatibility.
IX-052	Open for the literal ind-system	C.10 retains the construction in $\mathrm{Ind}(\mathrm{Hot}_*)$ and $\mathrm{Ind}(\mathrm{Cat}\,\mathrm{hot}_*)$ with all pointed transition coherences. The sufficient pointed enrichment of C.4 is not asserted to identify these particular ind-objects.
IX-061	Proved for the dg model; Open generally	C.8 constructs $\tau_{\le 0}\mathbf{R}\mathrm{Hom}_k(a,b) \in D^{\le 0}(k)$ for complexes and the stated dg enrichments. The unaxiomatized additive relative-category proposal, and its comparison with literal full path nerves, remain C.9.
IX-065	Proved for the dg model; Open generally	The derived-tensor composition and units, including truncation and the Koszul factor-order convention, are proved in C.8. Ordinary k-linearity of the localization alone is not the enhancement hypothesis; the broader claim is retained in C.9.
IX-070	Proved for the dg model; Open generally	C.8 supplies the strong bilinear composition in its specified derived setting. C.9 records the missing general enhancement and literal-path comparison, including the scope of the stronger additive model-category sufficient conditions.
IX-072	Known under pointed comparison; Open otherwise	C.5 identifies the group at the zero map with Quillen's group after the stated replacements. Applying this to the manuscript's pointed full path object additionally needs a comparison carrying that basepoint, or a specified basepoint path; C.10 retains that transfer problem.
Chapter IX: the tests and loop–suspension assertions		
IX-080	Refuted in the weak reading; Open in the stronger reading	D.1 distinguishes both formulations of (b). Weak equivalence to a Thomason-fibrant category is automatic, whereas restriction-invariance can fail by D.5. The suitably bounded Cat hot local-representability interpretation is retained in D.22, item 5. The unrestricted small-index restriction is D.K4.
IX-081	Refuted as a completeness criterion; precise criteria qualified	The finite example satisfies the tests and is rectifiable: D.2. Its missing finite products are explained by D.3, so those tests do not imply model or derivator completeness. D.15 gives the family-level Kaledin criterion on its actual shapes; the smaller-shape recognition questions remain D.22.
IX-084	Proved under hypotheses; weaker inference Refuted	C.6 and D.6 prove the pointed mapping-space identities when loops and suspension are the requisite homotopy pullback and pushout. C.7 refutes the inference from representing only π_1. Transfer to the exact pointed path model remains C.10.
IX-087	Proved for pointed derivators; conditional for a naive family	C.6 proves the derivator adjunction and preservation of homotopy limits by loops and homotopy colimits by suspension. D.7, D.17, and D.18 give sufficient diagram theories with their size hypotheses. No such structure is inferred merely from the notation $M^A[W_A^{-1}]$.

Continued on the following page.

Target	Disposition	Exact scope and numbered location
IX-088	Open for the literal implication; Proved in sufficient settings	The implication from the manuscript's path-object (c) to its restriction-invariance (b) remains D.22, item 5. D.4 proves rectification implies (b), and D.6 proves (c) implies the corresponding part of (a). These distinct implications do not establish the missing general one.
Chapters V and VII: sequential models and endpoint properties		
VIIc-A046	Refuted in Cat hot; Proved in Hot for the path systems	C.P5 gives a four-object crown for which the actual q_L has no inverse modulo finite zigzags of natural transformations. The specified cofinal type-system comparisons are weak equivalences by C.P3. A.9 separately checks the finite subdivision warning.
V-I066	Refuted in the stronger category; qualified positive comparison	For the specified cofinal projective type systems, canonical transitive comparison of unions in Hot is C.P3. Promotion to Cat hot is false by C.P5. Isomorphism of ind-objects alone does not supply either comparison; other path-model identifications retain B.8.
VIIb-C026	Refuted generally; Proved under the stated criterion and for paths	A.4 refutes preorder cofinality plus parallel-map homotopies for arbitrary fibrations. A.3 gives the exact homotopy-initiality criterion uniform over all diagrams. C.P1–C.P3 prove the stronger path-specific descent argument and the actual comparison roof of A.P2.
VIIc-C004	Translation Proved; faithfulness Refuted	E.1 proves that the literal even translation is the identity in Hot. E.13 independently verifies the finite-support counterexample to faithfulness of finite set-valued tests. The specified equality holds, while the proposed general inference from those tests is false.
VIIc-C005	Proved for the eventual model; all-type strengthening Refuted	For the eventual alternating endpoint functor, smoothness, properness, and the strict homotopy-fibre assertion are E.1 and E.4. The all-type endpoint functor is smooth but need not be proper or a Grothendieck (op)fibration, by E.3. These are different functors.
Numbered targets inherited from the earlier packages		
CATHOT E.26(1)	Proved	C.P3: for strictly commuting $u : t \to s$ between preorder-cofinal projective type sequences, the actual $u^* : C_\infty(s) \to C_\infty(t)$ is in W_∞, for every small M, wide Σ, and endpoint pair.
CATHOT E.26(2)	Proved	C.P3: the comparisons through $t * s$ equal $\beta_t^{-1}\beta_s$ in Hot. This proves independence and transitivity for the specified maps and roofs, rather than merely an abstract isomorphism of their sources and targets.
CATHOT E.26(3)	Proved	C.P3: the specified fibred Grothendieck functor $E_t \to \mathrm{Ch}_\Sigma(M; a, b)$ is in W_∞. A.P2 identifies its roof with the sequential colimit. The proof does not assume that J is a weak equivalence.

Continued on the following page.

Target	Disposition	Exact scope and numbered location
CATHOT E.27	Proved for every cofinal block	C.P4: for any block containing both signs, both actual padding maps are weak equivalences, and the resulting operation is associative, unital, and independent of the block in Hot. A one-sign block does not give a cofinal type system and is outside this assertion.
CATHOT E.17	Proved in the small-index scope	D.K1–D.K4 prove the acyclic-map reduction, the cardinal-controlled Kan–Thurston input, and factorization through Π_1 for a small filtered ind-system tested on all small categories in the same universe. D.K5 applies this to the countable path system and records the bounded-shape limitation.
STRATA E.11	Open; standard path constructions Refuted as solutions	E.6 proves that the eventual endpoint construction fails both strict lifting properties over the five-element rectangular-band monoid; the all-type construction fails the opfibration property. E.11 retains existence of a different strict bifibration realizing its path local system.
The three leads and the additional rectification targets		
Lead 1	Known restricted model; Open general inclusion	B.1 specifies J and B.2 identifies its restricted source with the hammock mapping space. B.3 confirms J need not be a categorical isomorphism, even with saturated markings. The general weak-equivalence question, including the model and Brown cases, remains B.8.
Lead 2	Proved, with a correction	B.4 proves the assertion whenever $\Sigma \subseteq \mathrm{Iso}(M)$, with a uniform contraction of the full category in Cat hot. The suggested initial object need not exist in the total category. For all arrows marked, J is the identity.
Lead 3	Proved sufficient implication; finite-example claim Refuted	D.7 proves the rectification statements with their actual finite-direct or universe-relative shape bounds; D.4 proves the second (b). The claim that the manuscript's finite example fails rectification is false by D.2: it rectifies every shape and satisfies Kaledin's axioms.
Finite-shape recognition and product tests	Refuted in the cubical form; Open in the stated generality	D.10 refutes the proposed cube criterion. D.21 upgrades the canonical comparison when all Kaledin product shapes are available. Necessary-and-sufficient criteria using only finite posets, finite categories, or a prescribed cardinal bound, and the finite-product-shape upgrade, remain D.22.
Intrinsic cofibrations and later axioms	Proved with the stated existence hypotheses; Open without them	D.8 proves finite rectification with an initial object and pushouts; D.9 refutes automatic cofibrancy of every object. D.17 compares XVII, XIX, and XIII and identifies the transfinite hypotheses for all-shape rectification. The version without an initial object remains D.22.
Barwick–Kan fibrancy; $B\mathbb{Z}$	Two implications Refuted; finite-shape implication Open	D.11 gives a Barwick–Kan fibrant example rectifying finite posets but not $B\mathbb{Z}$. D.12 gives an all-shape rectifiable example which is not fibrant. Whether fibrancy forces finite-poset or finite-category rectification remains D.22.

Continued on the following page.

Target	Disposition	Exact scope and numbered location
Relative poset presentations	Refuted as a rectification criterion	D.13 gives a relative poset presenting an ordinary category whose strict diagram family nevertheless fails the specified comparison. Presenting the correct localization does not by itself rectify diagram categories.
Kaledin recognition and extension	Known with actual shapes; naive extension obstruction Proved	D.14 audits the axioms and shape classes. D.15 identifies the object represented by a naive enhanced family as its specified localization. D.16 locates an explicit countable-poset failure of additivity; the finite example of D.2 has no failed enhanced-category axiom.
Accessible localizers; descent classes	Known with accessibility; broader assertion Open	D.18 gives all-shape rectification and enough intrinsic cofibrations for accessible localizers. D.23 proves that every nonzero spectrum's homology localizer lacks enough intrinsic fibrations. D.19 refutes two-out-of-three for the stated descent class. The nonaccessible derivator question remains D.22.
Recovery from diagram homotopy categories	Known with the full coherent family	D.20 resolves the Arlin and Fuentes-Keuthan–Kędziorek–Rovelli references and distinguishes their shape and morphism conventions. Its arrow-shape example refutes replacing coherent diagram homotopy categories by diagram categories in the homotopy category.

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